

Mechanism Design under Costly Signaling: the Value of Non-Coordination*

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Abstract

We study allocation mechanisms that utilize costly signaling as a screening tool. A social planner aims to maximize social welfare, defined as the weighted sum of agents' utilities, while implementing a specific allocation rule. Within a broad class of agent preferences, we show that coordination mechanisms (where recommended signals depend on joint reports) can be outperformed by non-coordination mechanisms (where signals depend solely on individual reports). We formalize the conditions under which the optimal mechanism features no coordination and demonstrate that such mechanisms are implementable through coarse-ranking contests.

Keywords— mechanism design, costly signaling, non-coordination, information leakage, coarse ranking contests.

JEL— D47, D61, D82

*We thank Asher Wolinsky, Bruno Strulovici, and Wojciech Olszewski for advice, helpful conversations, and comments. We also thank Ian Ball, Dirk Bergemann, Alex Bloedel, Simon Board, Eddie Dekel, Piotr Dworczak, Jeff Ely, Kira Goldner, Carl-Christian Groh, Yingni Guo, Marina Halac, Andrei Iakovlev, Annie Liang, Bart Lipman, Brendan Lucier, Deniz Kattwinkel, Joshua Mollner, Kyohei Okumura, Alessandro Pavan, Marcin Peški, Abhishek Sarkar, James Schummer, Philipp Strack, Matthew Thomas, Yiyao Zhu, and participants at the UCLA Theory Seminar, 34th Stony Brook International Conference on Game Theory, Australasian Economic Theory Week, NUS Theory Lunch for helpful suggestions, comments, and discussions. Yingkai Li acknowledges financial support from NUS Start-up Grant. Part of the work was done while Yingkai Li was a Postdoc at Yale University under the support of the Sloan Research Fellowship, grant no. FG-2019-12378.

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1 Introduction

Economic resources are often allocated based on informative but costly signals. When resources are owned by a central planner, the classic signaling framework of Spence (1973) no longer applies. Further, agents compete with excessive signaling effort, reducing social welfare. Therefore, we adopt a mechanism design framework and ask which mechanism maximizes welfare while implementing a given allocation rule.

While this question is central to the money-burning literature (e.g., Hartline and Roughgarden, 2008; Condorelli, 2012), we highlight a fundamental yet overlooked tradeoff. Because the principal can communicate with all agents, she may improve welfare by coordinating their signaling, for example by advising likely losers to refrain from effort. However, such coordination requires credibly identifying the winner, which may necessitate a more stringent, costly signal from that agent. Coordination thus reduces waste on one margin but increases it on another. The money-burning literature typically assumes quasi-linear preferences and does not consider this tradeoff, because the payoff equivalence result of Myerson (1981) implies that it is without loss of optimality to use *non-coordinated* (all-pay) mechanisms. In this paper, we study environments where payoff equivalence may not hold and ask: when is optimal welfare achievable via simpler, non-coordinated mechanisms?

We consider a broad class of preferences that captures heterogeneous valuations and diverse signaling technologies, including classic sorting conditions (Spence, 1973), designable costs (Chakravarty and Kaplan, 2013), and type-independent costs (Yang et al., 2024). A prominent class of signaling technologies covered by our model is one where agents can reveal their true type for free but must incur private costs to inflate their signal (Perez-Richet and Skreta, 2022). These signaling technologies encompass real-world frictions, such as lying cost in public housing allocation, where applicants either truthfully present financial evidence or exert effort to underreport income to meet eligibility thresholds (U.S. Government Accountability Office, 2005); head starts in college admissions, where students with greater talent or access to tutoring resources exert less effort to achieve the same standardized test scores as their peers; and financing costs in performance-based procurement, where a firm must pay to issue debt above its current cash holdings to bolster its competitive position (Che and Gale, 1998).

Our main result identifies a crucial condition for the optimality of non-coordinated mechanisms: Increasing Certainty Equivalence (ICE). ICE requires that higher types, being more efficient at signaling, are better able to generate a deterministic equivalent of any stochastic signal distribution. Under ICE, we show that welfare-maximizing mechanisms are always non-coordinated (Theorem 1).

Our result offers two insights. First, it serves as a robustness result: it generalizes the

classic optimality of all-pay implementations from quasi-linear settings, where payoff equivalence holds, to a much broader class of preferences. This suggests that this optimality is a fundamental feature of preference structures rather than a byproduct of payoff equivalence. Second, when a strict version of ICE holds, we show that coordination mechanisms are strictly sub-optimal. The cost of maintaining incentive compatibility in a coordination mechanism strictly exceeds the welfare saved by reducing waste. From this perspective, our results suggest that despite the potential for coordination to reduce wasted effort, letting agents compete publicly without designer intervention minimizes incentive costs and maximizes welfare. This provides a simplicity result for policymakers: in many realistic settings, there is no value in seeking elaborate direct mechanisms; optimal welfare is achieved through simple, non-coordinated competition.

The primary technical challenge in our analysis is comparing coordination and non-coordination mechanisms. Because payoff equivalence does not necessarily hold in our setting, coordination and non-coordination mechanisms implementing the same allocation rule can yield different utility levels for agents, even when the lowest type's utility is held constant. To analyze coordination mechanisms, we adopt a constructive approach. While a coordinating principal issues stochastic signal recommendations based on joint reports, a non-coordination mechanism relies on deterministic ones. We prove that for any stochastic mechanism, we can construct a deterministic counterpart that implements the same allocation while increasing every agent's utility under ICE. This construction demonstrates that the waste saved by coordination is always outweighed by the increased cost of ensuring truth-telling in a stochastic environment.

To illustrate how coordination impacts welfare and our core constructive idea, we compare a winner-pays mechanism (which coordinates signals based on joint reports) with an all-pay mechanism (which does not) in the following simplified setting (generalized in Section 2).

Example 1. A principal allocates a single indivisible item to two agents based on public signals. Each agent's private type θ_i , is drawn i.i.d. from a discrete uniform distribution on $\{L = 0, H = \frac{1}{2}\}$. Agent i can generate a signal $s_i \in [0, \infty)$ at zero cost if $s_i \leq \theta_i$; however, inflating the signal above their type ($s_i > \theta_i$) incurs a quadratic cost $c_i = \frac{1}{2}(s_i - \theta_i)^2$. With a common valuation of 1 for the item, an agent's utility is $u_i = x_i - c_i$, where x_i represents the allocation probability. The principal awards the item to the higher type, using a uniform tie-break. We compare two mechanisms that implement this efficient allocation rule.

Winner-pays (Coordinated) Mechanism: Each agent reports a type. Based on the reports, a potential winner is chosen as the one who reports a higher type under uniform tie-breaking. To maximize welfare, type L is recommended to signal 0. Moreover, to deter

type L from mimicking type H , type H is recommended to signal $\frac{2}{\sqrt{3}}$ *only if* they win.¹ The expected utility is $\frac{1}{4} \cdot (1 - 0) = \frac{1}{4}$ for type L and is $\frac{3}{4} \cdot (1 - \frac{1}{2}(\frac{2}{\sqrt{3}} - \frac{1}{2})^2) \approx 0.589$ for type H . Intuitively, the winning agent must burn utility to credibly signal superiority.

All-pay (Uncoordinated) Mechanism: Type L is recommended with a deterministic signal 0, and type H is recommended with a deterministic signal 1, regardless of whether they win the item or not.² The item is allocated to the type with a higher signal, with ties broken uniformly randomly. While the expected utility for type L remains unchanged, the expected utility for type H is $\frac{3}{4} - \frac{1}{2}(1 - \frac{1}{2})^2 = 0.625 > 0.589$, strictly higher than that in the winner-pays mechanism.

The key distinction between these two mechanisms is that recommendations are deterministic functions of each agent's own type in the all-pay mechanism, while in the winner-pays mechanism, they are stochastic from each agent's interim perspective, where the randomization arises from the randomization of others' reports. The latter coordinates signals so that only the winner expends effort, saving the loser's cost. However, this coordination raises incentive costs: the winner's signal must be sufficiently high to deter imitation.

To be more concrete, type H receives a stochastic signal recommendation in the winner-pays mechanism: with probability $\frac{3}{4}$, they are the potential winner with a signal requirement $\frac{2}{\sqrt{3}}$; with probability $\frac{1}{4}$, they are picked as the loser with no signal requirement. Let $\bar{s}$ and $\underline{s}$ be the deterministic signals that yield the same utility as this stochastic lottery for types H and L , respectively. Under ICE, which is satisfied by the quadratic cost, $\bar{s} > \underline{s}$.³

If we replace the winner-pays lottery with a deterministic signal $\bar{s}$ for type H , type H 's utility remains unchanged. However, because $\bar{s} > \underline{s}$, this deterministic signal is more costly for a low type to mimic than the original lottery was. This slackness in incentive compatibility allows the principal to lower the signal requirement for type H while still deterring type L . Consequently, the all-pay mechanism reduces the total waste required to maintain truth-telling, leading to higher social welfare.

The observation that removing coordination improves welfare is not contingent on the quadratic cost structure or efficient allocations. Rather, the fundamental requirement is the ICE condition. This condition guarantees that a non-coordination mechanism can mitigate

¹Under this construction, type H is strictly worse off by mimicking type L . We highlight that to maximize welfare, upward deviation is the binding constraint, whereas in the classical revenue-maximization auction design problems, downward deviation is the binding constraint. We will discuss this contrast in greater detail in Section 3.2.

²In this mechanism, type H is strictly worse off by mimicking type L , while the upward deviation incentive is binding: the payoff of type L from mimicking type H is $\frac{3}{4}(1 - \frac{1}{2}(1 - 0)^2) = \frac{1}{4}$, conditional on the other agent playing the equilibrium strategy.

³The high and low type's certainty equivalent signal $\bar{s}$ and $\underline{s}$ satisfy $\frac{1}{2}(\bar{s} - \frac{1}{2})^2 = \frac{3}{4} \cdot \frac{1}{2}(\frac{2}{\sqrt{3}} - \frac{1}{2})^2$ and $\frac{1}{2}(\underline{s} - 0)^2 = \frac{3}{4} \cdot \frac{1}{2}(\frac{2}{\sqrt{3}} - 0)^2$, respectively. We therefore have $\bar{s} = \frac{3}{2} - \frac{\sqrt{3}}{4} \approx 1.06$ and $\underline{s} = 1$.

(upward) deviation incentives, which lead to welfare-improvement. Our main theorem (Theorem 1) formalizes this intuition by demonstrating that for any *monotone* interim allocation, coordinated signal recommendations are dominated by non-coordination mechanisms. While non-monotone allocation rules are often infeasible in practice due to fairness concerns, we establish in Section 5 that even when such allocations are implementable, they are generally dominated by monotone allocations under mild conditions.

Our central contribution is the observation that non-coordination can be optimal in environments where payoff equivalence fails. Although Perez-Richet and Skreta (2024) study a similar quadratic cost structure as in our Example 1, their analysis is restricted to a single-agent model, thus bypassing the coordination problem entirely. Our findings also depart from the money-burning literature, which typically relies on quasi-linear preferences to ensure payoff equivalence. By relaxing these assumptions, we identify a novel economic tradeoff inherent in coordination. This framework allows us to address a new design question: whether complex coordination mechanisms are necessary for welfare maximization. Finally, while Zhang (2024) demonstrates the optimality of all-pay mechanisms for effort maximization under separable convex costs, our model focuses on welfare and accommodates a broader class of type-dependent and non-convex cost functions.

The remainder of the paper is organized as follows. In the next subsection, we relate our work to the existing literature. In Section 2, we introduce the general model. In Section 3, we present our main result and the intuition behind it. In Section 4, we illustrate the connections between non-coordination (all-pay) mechanisms and traditional ranking-based contests by proposing coarse ranking contests as practical implementations of non-coordination mechanisms in symmetric environments. In Section 5, we present a partial converse result on when coordination can improve welfare, and generalize the optimality of non-coordination mechanisms to non-monotone allocations. In Section 6, we conclude with directions for future research.

1.1 Related Work

Our paper is closely related to the literature on money burning and costly ordeals. Prior work in this area typically assumes that agents' preferences are quasi-linear in transfers (Hartline and Roughgarden, 2008; Condorelli, 2012; Finkelstein and Notowidigdo, 2019) or in effort-based costs independent of private information (Chakravarty and Kaplan, 2013; Yang et al., 2024). Under such restrictions, the mechanism design space simplifies: for any mechanism that coordinates agents' signaling choices, there exists a non-coordinating counterpart that yields identical utilities for each agent. This invariance suggests that coordination offers no inherent welfare advantage in standard models. This logic traces back to McAfee and

McMillan (1992), who studied bidding rings in first-price auctions. In their setting, if a “weak cartel” cannot use side-payments, the optimal collusive strategy is simply to have all members submit identical bids at the reserve price, effectively a non-coordinated outcome. When signaling costs are identical and type-independent, our environment is conceptually similar to theirs; the bids paid to a third-party seller act as the wasteful signals that serve as screening devices in our model. A key distinction is that while McAfee and McMillan (1992) focus on collusion between agents, the coordination in our setting is performed by the principal, who integrates reports from all agents to determine signaling requirements.

We depart from this literature by relaxing the quasi-linearity assumption. In our more general setting, mechanisms implementing the same allocation rule can result in disparate welfare outcomes. While this expansion of the feasible mechanism space introduces new technical challenges, it allows us to uncover a novel economic tradeoff regarding the optimality of coordination. We introduce the Increasing Certainty Equivalence (ICE) assumption (Assumption 3), under which we prove that welfare-maximizing mechanisms feature no coordination. We thus provide a robust generalization of the optimality of non-coordination mechanisms to a broader class of preferences.

Our paper is also related to mechanism design beyond quasi-linearity, where transfers generate wealth effects through risk aversion or financial constraints. In such environments, payoff-equivalence arguments can fail and the details of implementation can matter for welfare and incentives (e.g., Maskin and Riley, 1984; Matthews, 1987; Che and Gale, 1998). More broadly, recent work studies incentive compatibility and mechanism design without quasi-linearity under general nonlinear preferences (e.g., Kos and Messner, 2013; Kazumura et al., 2020). Our setting differs in that monetary transfers are replaced by costly (possibly type-dependent) signals, and the principal seeks to maximize agents’ utilities. Nonetheless, the shared theme is that nonlinearity enlarges the relevant design space; our ICE condition restores tractability by implying that non-coordination mechanisms weakly dominate coordinating ones for implementing any monotone allocation.

Our paper is also conceptually related to the literature on falsification and costly lying. Green and Laffont (1986) study auction settings where lying costs are either zero or infinite, while later papers consider richer cost structures but focus on different objectives, such as classification (Hardt et al., 2016) or test design (Perez-Richet and Skreta, 2022). The most closely related work is Perez-Richet and Skreta (2024). They show that under linear or quadratic falsification costs, the optimal mechanism is score-based; that is, it recommends a deterministic action based on the report. The main difference is that they study a single-agent model, ruling out the possibility of coordination. In contrast, we study a multi-agent environment. This allows us to study the welfare consequences of coordinating agents’

signaling choices.

The costly signaling aspect of our model is rooted in the classic signaling literature (e.g., Spence, 1973) and the more recent work on strategic gaming of signals (e.g., Frankel and Kartik, 2019; Ball, 2024). The main difference is that, unlike standard signaling games where markets reward agents based on beliefs about their type, we study institutional settings where a principal directly controls allocation and can design mechanisms under resource constraints. This design perspective enables us to analyze how different implementations of the same allocation rule lead to different agent behaviors and social costs.

Our non-coordination mechanisms can also be implemented as coarse ranking contests, which generalize traditional contest formats such as all-pay auctions (Baye et al., 1996), tournaments (Lazear and Rosen, 1981), and Tullock contests (Skaperdas, 1996). Coarse rankings allow the mechanism to map multiple performances into the same tier, enabling robustness and reducing unnecessary effort. Zhang (2024) also shows that contests can be optimal for inducing effort, under special signaling costs. Our key departure from this literature is that we focus on minimizing socially wasteful effort.

2 Model

The principal (she) wishes to allocate k identical items to $n > k$ heterogeneous agents (he). An allocation $\mathbf{x} = (x_i)_{i=1}^n$ is a vector of probabilities such that $0 \leq x_i \leq 1$ for each i , and $\sum_{i=1}^n x_i \leq k$.⁴ Let $X \subseteq [0, 1]^n$ be the space of feasible allocations.

Each agent i has a private type θ_i drawn independently from a publicly known distribution F_i supported on $\Theta_i = [\underline{\theta}_i, \bar{\theta}_i] \subseteq \mathbb{R}_+$. The distribution of the type profile $\boldsymbol{\theta}$ is $\mathbf{F}$, whose support is $\boldsymbol{\Theta} = \prod_{i=1}^n \Theta_i$. Each agent i can generate a costly public signal $s_i \in S = \mathbb{R}_+$ and the utility of the agent is $u_i(x_i, s_i, \theta_i)$ when receiving an item with probability $x_i \in [0, 1]$ and generating a signal s_i . We assume that $u_i(x_i, 0, \theta_i) \in [0, 1]$ for any x_i, θ_i . That is, all agents have a weakly positive value for any allocation given their types when not generating any costly signal, and this value is bounded. Denote the Cartesian product of agents' signal space by $\mathbf{S} = S^n$. The principal cannot observe agents' private types, but she can observe their signals.

Assumption 1 (regularity). For any agent i , his utility function u_i is continuous in all of its coordinates, bounded in value, strictly increasing in x_i , weakly decreasing in s_i , and

⁴Our paper focuses on the interpretation where items are indivisible and x_i represents allocation probabilities. All the results extend naturally by considering divisible items and interpreting x_i as fractional allocations.

weakly increasing in θ_i . Moreover, it satisfies the von Neumann-Morgenstern expected utility representation.

Under Assumption 1, each agent always strictly prefers a higher probability of allocation. Moreover, since it is costly to generate any signals, the cost of signaling is a weakly increasing function of signal production. There are two potential reasons why the utility is an increasing function of type. First, a higher type may value the item more. Second, a higher type may have a weakly lower cost of generating the same signal. Our assumption is general enough to cover both scenarios.

Mechanisms. We want to define a class of mechanisms as general as possible. For example, we want to incorporate the possibility of communication between the principal and agents, and the choice of the mechanisms can be contingent on such communication. Following Myerson (1982), we know that it is both convenient and without loss of generality to study direct mechanisms.⁵ In these direct mechanisms, we allow the principal to receive type reports from all agents, based on which she makes signal recommendations. Moreover, the allocation of items can be contingent on both the reports and the observed signals.

Formally, the principal first commits to a signal recommendation policy $\tilde{s} : \Theta \rightarrow \Delta(\mathcal{S})$ and an allocation rule $\mathbf{y} : \Theta \times \mathcal{S} \rightarrow X$. Then each agent i reports $\hat{\theta}_i$ to the principal and receives a signal recommendation $\tilde{s}_i(\hat{\theta})$. Based on the recommendation, each agent i produces signal $s'_i \in \{\tilde{s}_i(\hat{\theta}), 0\}$.⁶ In the end, the principal observes the signal profile $\mathbf{s}'$. Each agent i receives an item with probability $y_i(\hat{\theta}, \mathbf{s}')$. We denote this direct mechanism by $(\tilde{s}, \mathbf{y})$.

We highlight a special class of direct mechanisms that will be of importance in our analysis. These are direct *non-coordination* mechanisms. They impose the following additional restrictions on direct mechanisms: (1) the signal recommendation policy for each agent i is a function of his type, that is, $\tilde{s}_i : \Theta_i \rightarrow S_i$, and (2) the allocation rule can only depend on public signals, that is, $\mathbf{y} : \mathcal{S} \rightarrow X$. These restrictions rule out the possibility of using reports from all agents to coordinate agents' choices of signals.

⁵In Myerson (1982), such mechanisms are called direct coordination mechanisms. We call them direct mechanisms or general mechanisms, because some of them exhibit zero coordination.

⁶We impose interim participation constraints for agents, i.e., agents can choose to walk away after seeing the signal recommendation. Our formulation is without loss of generality under the interim participation constraints since the principal can partially enforce the recommendation by allocating no items to any agent who chooses a signal different from the recommended one. Here each agent essentially has two choices: following the recommendation ($s'_i = \tilde{s}_i(\hat{\theta})$) or opting out ($s'_i = 0$). An alternative formulation is to only consider the ex ante participation constraint where each agent only has the option to opt out at the beginning of the mechanism, and they are forced to follow the principal's recommendation after participation. This distinction is not crucial for our analysis, and the main results of the paper hold for both cases.

It is worth noting that direct non-coordination mechanisms can be implemented as *signal-based* mechanisms. In a signal-based mechanism, the principal commits to a signal-based allocation rule $\mathbf{x} : \mathbf{S} \rightarrow X$, which maps each realized signal profile to a (randomized) allocation. After that, each agent i chooses a signal strategy $s_i : \Theta_i \rightarrow \Delta(S)$ to maximize his expected utility. Given the strategy profile $\mathbf{s} = (s_i)_{i=1}^n$ and the realized signal profile $\mathbf{s}(\boldsymbol{\theta}) = (s_i(\theta_i))_{i=1}^n$, an item is allocated to agent i with probability $x_i(\mathbf{s}(\boldsymbol{\theta}))$. For any direct non-coordination mechanism, it is straightforward to show that there exists a signal-based mechanism that implements the same outcome under a pure Nash equilibrium.⁷

Interim approach. Given any direct mechanism $(\tilde{\mathbf{s}}, \mathbf{y})$, the *interim allocation* and the *interim utility* of agent i with private type θ_i is

$$Q_i(\theta_i) = \mathbf{E}_{\boldsymbol{\theta}_{-i}} [\mathbf{E}_{\mathbf{s} \sim \tilde{\mathbf{s}}(\theta_i, \boldsymbol{\theta}_{-i})} [y_i(\theta_i, \boldsymbol{\theta}_{-i}, \mathbf{s})]] \quad (1)$$

and

$$U_i(\theta_i) = \mathbf{E}_{\boldsymbol{\theta}_{-i}} [\mathbf{E}_{\mathbf{s} \sim \tilde{\mathbf{s}}(\theta_i, \boldsymbol{\theta}_{-i})} [u_i(y_i(\theta_i, \boldsymbol{\theta}_{-i}, \mathbf{s}), s_i, \theta_i)]] \quad (2)$$

respectively.

A direct mechanism $(\tilde{\mathbf{s}}, \mathbf{y})$ is incentive compatible (IC) if, for any agent i and any types θ_i, θ'_i , his expected utility from truthfully reporting his type and following the signal recommendation is not lower than that from misreporting as θ'_i and deviating from the signal recommendation $s_i = \tilde{s}_i(\theta'_i, \boldsymbol{\theta}_{-i})$ according to any function $s'_i(s_i) \in \{0, s_i\}$. That is,

$$U_i(\theta_i) \geq \mathbf{E}_{\boldsymbol{\theta}_{-i}} [\mathbf{E}_{\mathbf{s} \sim \tilde{\mathbf{s}}(\theta'_i, \boldsymbol{\theta}_{-i})} [u_i(y_i(\theta'_i, \boldsymbol{\theta}_{-i}, s'_i(s_i), \mathbf{s}_{-i}), s'_i(s_i), \theta_i)]]]. \quad (\text{IC})$$

We denote the interim allocation profile and interim utility profile as $\mathbf{Q} = (Q_i)_{i=1}^n$ and $\mathbf{U} = (U_i)_{i=1}^n$, respectively.

Definition 1 (implementability). An interim allocation–utility profile $(\mathbf{Q}, \mathbf{U})$ can be *implemented by a direct mechanism* if there exists a signal recommendation policy $\tilde{\mathbf{s}} : \boldsymbol{\Theta} \rightarrow \Delta(\mathbf{S})$ and an allocation rule $\mathbf{y} : \boldsymbol{\Theta} \times \mathbf{S} \rightarrow X$ such that (i) the direct mechanism $(\tilde{\mathbf{s}}, \mathbf{y})$ is incentive compatible, and (ii) the consistency conditions (1) and (2) hold for any agent i and type θ_i .

An interim allocation $\mathbf{Q}$ can be *implemented* (by a direct mechanism) if there exists an interim utility profile $\mathbf{U}$ such that $(\mathbf{Q}, \mathbf{U})$ can be implemented by a direct mechanism.

Analogously, an interim allocation–utility profile $(\mathbf{Q}, \mathbf{U})$ can be *implemented by a direct*

⁷A similar observation on this indirect implementation is also made in the contemporary work by Perez-Richet and Skreta (2024) in a single-agent environment.

non-coordination mechanism if the direct mechanism $(\tilde{\mathbf{s}}, \mathbf{y})$ in Definition 1 is a direct non-coordination mechanism. That is, the signal recommendation policy for each agent i only depends on his own type: $\tilde{s}_i : \Theta_i \rightarrow S$, and the allocation rule only depends on the public signals: $\mathbf{y} : \mathbf{S} \rightarrow X$. The consistency condition now becomes

$$\begin{aligned} Q_i(\theta_i) &= \mathbf{E}_{\boldsymbol{\theta}_{-i}}[y_i(\tilde{s}_i(\theta_i), \tilde{\mathbf{s}}_{-i}(\boldsymbol{\theta}_{-i}))], \\ U_i(\theta_i) &= \mathbf{E}_{\boldsymbol{\theta}_{-i}}[u_i(y_i(\tilde{s}_i(\theta_i), \tilde{\mathbf{s}}_{-i}(\boldsymbol{\theta}_{-i})), \tilde{s}_i(\theta_i), \theta_i)]. \end{aligned} \tag{C'}$$

Notice that in a direct mechanism, both the allocation and the recommended signal could be stochastic from each agent's perspective, whereas in a direct non-coordination mechanism, only the allocation could be stochastic from each agent's perspective. We introduce two more simplifying notations. For any agent i , denote the expected utility of type θ by $U_i(\theta; G) := \mathbf{E}_{(x_i, s_i) \sim G}[u_i(x_i, s_i, \theta_i)]$ when the allocations and recommended signals follow the distribution G . Denote the expected utility of type θ by $U_i(\theta; G_X, s)$ when the allocations follow the distribution G_X and the signal is $s \in \mathbb{R}_+$.

Principal's payoff. The principal is a social planner who cares about maximizing the utilities of agents when implementing any targeted allocation rule. The principal relies on public signals to screen agents, but is aware that excessive signaling is socially wasteful. Formally, given any allocation rule $\mathbf{Q}$ that can be implemented by a direct mechanism, the main objective is to design a mechanism that implements $\mathbf{Q}$ and maximizes the weighted utilities of the agents, that is,

$$\sum_{i=1}^n \mathbf{E}_{\theta_i \sim F_i}[w_i(\theta_i) \cdot U_i(\theta_i)],$$

where $w_i(\theta_i) \in [0, 1]$ is a weighting function that can depend on each agent's private type.

3 Optimality of Non-coordination Mechanisms

In this section, we show that under some conditions, given any monotone allocation rule, the optimal mechanism is a non-coordination mechanism.

3.1 Assumptions

We first introduce the standard weak single-crossing property as in Milgrom and Shannon (1994).

Assumption 2 (weak single-crossing property (wSCP)). For any agent i , his utility function u_i satisfies *weak single-crossing property (wSCP)*: for any types $\theta'_i > \theta_i$, any signals $s'_i \geq s_i$ and any allocation probabilities $x'_i \geq x_i$,

1. $u_i(x'_i, s'_i, \theta_i) - u_i(x_i, s_i, \theta_i) > 0$ implies that $u_i(x'_i, s'_i, \theta'_i) - u_i(x_i, s_i, \theta'_i) > 0$;
2. $u_i(x'_i, s'_i, \theta_i) - u_i(x_i, s_i, \theta_i) \geq 0$ implies that $u_i(x'_i, s'_i, \theta'_i) - u_i(x_i, s_i, \theta'_i) \geq 0$.

Note that Assumption 2 only requires comparisons across deterministic outcomes. This restriction is important in our setting because, in a general direct mechanism, an agent may face stochastic signal recommendations, especially when the principal coordinates agents' signaling choices using cross-agent information. As a result, Assumption 2 alone is typically not sufficient to ensure that local incentive constraints characterize global incentive compatibility in arbitrary direct mechanisms.⁸ To circumvent this technical challenge, we adopt a constructive approach: starting from any direct mechanism that implements a monotone allocation rule, we explicitly construct a non-coordination mechanism that implements the same allocation rule and weakly improves agents' utilities. This construction relies crucially on the following assumption.

Definition 2. Suppose allocations and signals follow the distribution G . Denote the marginal distribution over allocations by G_X . The certainty equivalent signal $\Sigma_i(G, \theta)$ of the distribution G for type θ satisfies the following conditions: $U_i(\theta; G_X, \Sigma_i(G, \theta)) = U_i(\theta; G)$.

Assumption 3 (Increasing Certainty Equivalence). A utility function u_i is said to have increasing certainty equivalence if for any distribution G over allocations and signals, his certainty equivalent signal $\Sigma_i(G, \theta)$ is increasing in θ .

Assumption 3 requires that higher-type agents are able to produce higher certainty-equivalent signals for any given lottery over signal recommendations. To illustrate this condition more concretely, consider a class of utility functions that are additively separable in valuation for the allocation and signaling cost

$$u_i(x_i, s_i, \theta_i) = v_i(x_i, \theta_i) - c_i(s_i, \theta_i).$$

Here, $v_i(\cdot, \cdot)$ represents the agent's valuation and $c_i(\cdot, \cdot)$ denotes the cost of signaling.⁹

⁸Kartik et al. (2024) study single-crossing differences (SCD), which impose single-crossing comparisons for arbitrary lotteries. This requirement is stronger than Assumption 2 and therefore applies to a narrower class of utility functions. For example, $u(x, s, \theta) = x - (s - \theta)^+$ satisfies Assumption 2 but not SCD.

⁹In settings without additive separability, we do not have simple sufficient conditions for Assumption 3. While we focus on additive separability for interpretability, our results apply whenever Assumption 3 holds. In Section E, we extend the core logic of Theorem 1 to non-separable utilities by proposing a weaker version of Assumption 3 and providing sufficient conditions for its validity.

Type-independent and Multiplicatively Separable Costs In this case, the utility function of the agents takes the form of $u_i(x_i, s_i, \theta_i) = v_i(x_i, \theta_i) - c_i(s_i)$. This includes the application of mechanism design with money burning or costly ordeals, where the agent's utility function typically takes the form $u_i(x_i, s_i, \theta_i) = \theta_i \cdot x_i - s_i$ (e.g., Hartline and Roughgarden, 2008). Moreover, this class of preferences can accommodate the following cases studied in the contest literature: $c_i(\cdot)$ is convex (e.g., Fang et al., 2020; Zhang, 2024), concave (e.g., Moldovanu and Sela, 2001), convex-then-concave or concave-then-convex (e.g., Kim et al., 2024), or even more complex structures.

More generally, our framework accommodates multiplicatively separable costs of the form: $c_i(s_i, \theta_i) = h_i(\theta_i) \cdot g_i(s_i)$. An agent with these preferences behaves identically to one with type-independent costs under the normalized utility $\frac{v_i(x_i, \theta_i)}{h_i(\theta_i)} - g_i(s_i)$. This allows us to capture settings in the redistribution literature where the agent's private type encodes two-dimensional information (valuation v_i and marginal cost c_i) and utility is $u_i = v_i \cdot x_i - c_i \cdot s_i$ (Akbarpour et al., 2024; Yang et al., 2024).

In these environments, Assumption 3 holds as a boundary case: the certainty-equivalent signal is constant across types because the conversion of a signal distribution into its deterministic equivalent depends solely on $c_i(s_i)$. In these cases, payoff equivalence holds (Myerson, 1981; Zhang, 2024): Any two mechanisms implementing the same interim allocation induce the same utilities for the agents. Thus, the optimality of non-coordination follows trivially.

Type-dependent Costs Our framework also accommodates genuinely type-dependent cost functions where payoff equivalence fails. While the classic signaling literature typically relies on the single-crossing property (Spence, 1973), Assumption 3 imposes additional structure on the curvature of the cost function.

Technically, Assumption 3 is equivalent to assuming that the disutility of signaling, $-c_i(s_i, \theta_i)$, exhibits weakly decreasing absolute risk aversion (DARA) in type θ_i for the cost of signaling (Pratt, 1976; Maskin and Riley, 1984). If $c_i(\cdot, \cdot)$ is third-differentiable, Assumption 3 is satisfied if and only if ¹⁰

$$\frac{\partial}{\partial \theta_i} \left(-\frac{c_{ss}(s_i, \theta_i)}{c_s(s_i, \theta_i)} \right) \leq 0, \quad \forall s_i, \theta_i, \quad \text{where } c_{ss}(s_i, \theta_i) = \frac{\partial^2 c_i(s_i, \theta_i)}{\partial s_i^2}, c_s(s_i, \theta_i) = \frac{\partial c_i(s_i, \theta_i)}{\partial s_i}. \quad (3)$$

Maskin and Riley (1984) provides further simplifying assumptions on primitives (in their assumptions A and B) such that the above condition holds.

¹⁰In Pratt (1976); Maskin and Riley (1984), one often additionally assumes risk aversion, i.e., $-c_{ss} \leq 0$. We do not require this restriction for Assumption 3.

While this connection is technically interesting, we highlight that ICE should be interpreted as a curvature restriction on signaling costs rather than a restriction on risk preferences over outcomes. Notice that we usually define “risk” over outcomes: $v_i - c_i$ if agent i gets an item and $-c_i$ if he does not get an item. Since preferences are linear in the allocation, agents remain risk-neutral in the traditional sense. ICE simply implies that higher types are more capable of generating higher deterministic equivalent signals to resolve the randomness in recommendations.

Type-dependent cost functions satisfying Assumption 3 arise naturally in several distinct economic contexts. In the costly falsification literature, $c_i(s_i, \theta_i)$ represents the effort required to inflate a report or score away from the truth. A commonly used reduced-form specification assumes that only upward falsification is costly, taking the form $c_i(s_i, \theta_i) = C((s_i - \theta_i)^+)$, for some increasing function $C(\cdot)$ with $C(0) = 0$ (Perez-Richet and Skreta, 2023). This formulation captures the intuition that the agent pays only for the incremental “gap” between the submitted signal and the true type. It nests several parametric forms used in applications, such as the linear cost $c_i(s_i, \theta_i) = (s_i - \theta_i)^+$ and the quadratic cost $c_i(s_i, \theta_i) = (s_i - \theta_i)^2$ are studied in Perez-Richet and Skreta (2024).¹¹

Similar distance-based costs appear in contest models with “head starts”, where a baseline advantage shifts the feasible frontier of attainable scores. These head starts may reflect innate talent, signaling greater merit, or socioeconomic advantages, such as access to elite tutoring. This structure is explored in Siegel (2009), where each player i chooses a score $s_i \in [a_i, \infty)$, and $a_i \geq 0$ is an initial score capturing the head start. By interpreting $e_i := s_i - a_i \geq 0$ as incremental effort, the cost representation $c_i(s_i, a_i) = C((s_i - a_i)^+)$, which aligns directly with our framework.

Finally, Assumption 3 can be interpreted as the cost of payments under financial constraints. In this context, it is useful to distinguish the baseline payment from the premium incurred when expenditures exceed available liquid funds. Che and Gale (1998) model the cost of paying x given budget w via a cost-of-spending function $C(x, w)$, and study the cost function $C(x, w) = x + R((x - w)^+)$ where $R(\cdot)$ represents an increasing, convex borrowing or financing premium satisfying $R(0) = 0$. Mapping the payment x to our signal s_i and the budget w to our type θ_i , we get a cost function

$$c_i(s_i, \theta_i) = s_i + R((s_i - \theta_i)^+).$$

Applying Pratt (1976), this specification satisfies Assumption 3 if $\log(1 + R'(z))$ is concave,

¹¹While the quadratic cost is not globally monotone in s_i , Perez-Richet and Skreta (2024) demonstrate that this is inconsequential for implementing monotone allocations, as equilibrium signal recommendations necessarily exceed the true type.

a condition met by both linear $R(z) = rz$ and the quadratic financial premia $R(z) = \kappa z^2$.

3.2 Main Theorem

Now we are ready to state our main theorem.

Theorem 1 (Optimality of Non-coordination). *Suppose Assumption 1, 2 and 3 hold. Fix any interim allocation-utility pair $(\mathbf{Q}, \mathbf{U})$ such that $(\mathbf{Q}, \mathbf{U})$ can be implemented by a direct mechanism and Q_i is weakly increasing for all i . Then there exists an interim allocation-utility pair $(\mathbf{Q}^\dagger, \mathbf{U}^\dagger)$ that can be implemented by a direct non-coordination mechanism such that $\mathbf{Q}^\dagger = \mathbf{Q}$, and $U_i^\dagger(\theta_i) \geq U_i(\theta_i)$ for all i, θ_i .*

We highlight that Theorem 1 can be strengthened to a strict version. Consider the following *strict* version of ICE: for any *non-degenerate* distribution G over allocations and signals, the certainty equivalent signal $\Sigma_i(G, \theta)$ is strictly increasing in θ .¹² Under this strict ICE condition, any coordinating mechanism that involves stochastic signal recommendations for some types is *strictly* worse than the optimal non-coordination mechanism.

Theorem 1 delivers a sharp simplification: under ICE, for any implementable *monotone* interim allocation, coordination in signal recommendations is never welfare-improving. Under the strict ICE condition, coordination is strictly sub-optimal. This suggests that to maximize agents' utilities, there is no need to utilize cross-agent information to coordinate signaling choice (or "leak" others' information to each agent). Coordination mechanisms typically involve stochastic signal recommendations, whose realizations depend on all agents' private information. Under the ICE condition, such stochastic recommendations can be improved upon using deterministic signals, where each agent's signaling choice does not depend on others' private information. This result suggests that the designer can safely use simple mechanisms like the non-coordination ones without sacrificing welfare.

A non-coordination mechanism admits an **all-pay** format. From this perspective, Theorem 1 suggests that the welfare-maximizing implementation admits an all-pay format when using costly signals as screening devices in allocation problems. At first glance, this may seem counterintuitive since all-pay formats require both winners and losers to incur signaling costs. However, we will show that all-pay formats can deter lower types from mimicking higher types more effectively. On average, each agent is required a "lower signal" while maintaining incentive compatibility. This allows every type to retain more information rent, improving utilities across the board. Overall, all-pay formats have a lower "incentive cost".

¹²The strict ICE holds, for example, when the condition in Eq. (3) holds with strict inequalities.

Proof Sketch. The proof is constructive and is provided in Section A. Here, we only outline the main steps. Consider an arbitrary agent i and an arbitrary type $\hat{\theta}_i$. Fix an arbitrary direct mechanism $\mathcal{M}$ that induces the interim allocation and interim utilities (Q, U) , where the allocation and recommended signal for type $\hat{\theta}_i$ follow a distribution G . We focus on the case where the distribution G is non-degenerate so that $\mathcal{M}$ coordinates agents' signals. To simplify the exposition, consider a finite type space and let $\tilde{\theta}_i$ be the largest type below $\hat{\theta}_i$.

Consider a candidate mechanism $\mathcal{M}^\dagger$ in which type $\hat{\theta}_i$ is recommended with a deterministic signal $\hat{s}_i$ and receives the item following the marginal distribution G_X induced by G , leaving the rest of the mechanism unchanged from the original mechanism. Let $U^\dagger$ be the interim utilities of the agents in mechanism $\mathcal{M}^\dagger$. We choose $\hat{s}_i$ so that the upward-deviation constraint of the lower type $\tilde{\theta}_i$ binds under the candidate mechanism, i.e., $U_i^\dagger(\tilde{\theta}_i) = U_i^\dagger(\tilde{\theta}_i; G_X, \hat{s}_i)$. Because the required signal is independent of the reports of other agents, the signaling choice of type $\hat{\theta}_i$ is uncoordinated. We want to show (1) **utility improvement**: the candidate mechanism improves interim utilities by removing coordination, and (2) **incentive compatibility**: the construction is valid in the sense that the candidate mechanism is incentive-compatible.

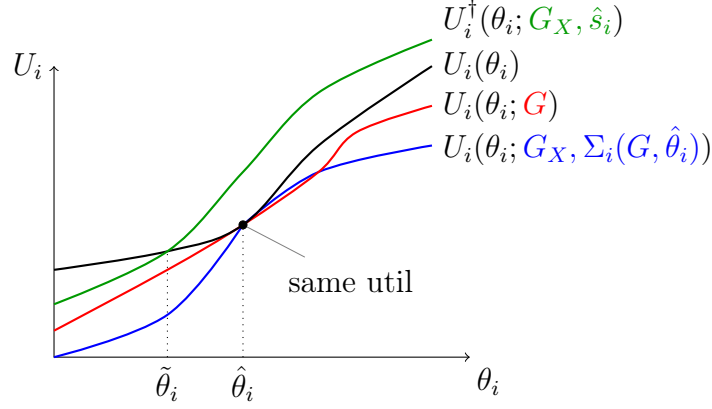
Utility improvement. To show (1), we only need to show that the interim utility of type $\hat{\theta}_i$ increases, since the rest of the mechanism remains the same as the original one. Thus, it suffices to show that the required signal $\hat{s}_i$ is weakly lower than type $\hat{\theta}_i$'s certainty-equivalent signal under the original mechanism, $\Sigma_i(G, \hat{\theta}_i)$.

To see this, consider an intermediary mechanism $\widehat{\mathcal{M}}$ in which type $\hat{\theta}_i$ is recommended with a deterministic signal $\Sigma_i(G, \hat{\theta}_i)$ and receives the item following the marginal distribution G_X , leaving the rest of the mechanism the same as the original mechanism $\mathcal{M}$. By construction, the interim utility, the expected utility of every type from truthful reporting, is the same as that in the original mechanism $\mathcal{M}$.¹³ Hence, we also denote the interim utility in $\widehat{\mathcal{M}}$ by $U_i(\theta_i)$. Moreover, the intermediary mechanism $\widehat{\mathcal{M}}$ differs from our candidate mechanism $\mathcal{M}^\dagger$ only in the deterministic signal recommended for type $\hat{\theta}_i$.

In addition, we introduce the deviation utilities, each type θ_i 's expected utility from mimicking $\hat{\theta}_i$, in all three mechanisms that will be helpful for the analysis: (a) in the original mechanism $\mathcal{M}$: $U_i(\theta_i; G)$, which equals to $U_i(\theta_i; G_X, \Sigma_i(G, \theta_i))$, (b) in the candidate mechanism $\mathcal{M}^\dagger$: $U_i^\dagger(\theta_i; G_X, \hat{s}_i)$, and (c) in the intermediary mechanism $\widehat{\mathcal{M}}$: $U_i(\theta_i; G_X, \Sigma_i(G, \hat{\theta}_i))$. These three deviation utilities correspond to the red, green, and blue curves in Figure 1, respectively. In addition, the black curve represents $U_i(\theta_i)$, the interim utility in $\mathcal{M}$ and $\widehat{\mathcal{M}}$.

¹³We highlight that the main purpose of introducing the intermediary mechanism $\widehat{\mathcal{M}}$ is to show (1). The incentive compatibility issue of the intermediary mechanism $\widehat{\mathcal{M}}$ does not affect our analysis, and so we will skip its discussion.

Figure 1: Comparison of Incentive Cost: Coordination vs Non-coordination



[†]The black curve is the interim utility of the agents in the original mechanism $\mathcal{M}$. The red curve captures the deviation utility for each type by mimicking type $\hat{\theta}_i$ in the original coordination mechanism $\mathcal{M}$. The blue curve captures the deviation utility for each type by mimicking type $\hat{\theta}_i$ in the intermediary mechanism $\widehat{\mathcal{M}}$. The green curve captures the deviation utility for each type by mimicking type $\hat{\theta}_i$ in the non-coordinated mechanism $\mathcal{M}^\dagger$.

We argue that by replacing the stochastic signal recommendation in the original mechanism $\mathcal{M}$ with a deterministic one in the intermediary mechanism $\widehat{\mathcal{M}}$, we relax the upward deviation incentive of the lower type $\tilde{\theta}_i$. This is implied by combining the observations that (i) the interim utility of type $\tilde{\theta}_i$ remains unchanged in all three mechanisms, and (ii) type $\tilde{\theta}_i$'s deviation utility from mimicking type $\hat{\theta}_i$ decreases in the intermediary mechanism $\widehat{\mathcal{M}}$. The latter claim is true because the ICE condition implies that $\Sigma_i(G, \theta_i) \leq \Sigma_i(G, \hat{\theta}_i)$ for all $\theta_i \leq \hat{\theta}_i$. Visually, this means that the red curve lies above the blue curve for all types below $\hat{\theta}_i$.¹⁴ Combining (i) and (ii), we conclude that the upward-deviation incentive of type $\tilde{\theta}_i$ is relaxed in $\widehat{\mathcal{M}}$ relative to $\mathcal{M}$. Intuitively, this slackness allows us to further reduce the deterministic signal recommended to type $\hat{\theta}_i$.

To make this intuition concrete, compare the candidate mechanism $\mathcal{M}^\dagger$ with the intermediary mechanism $\widehat{\mathcal{M}}$. Recall that the candidate mechanism $\mathcal{M}^\dagger$ is constructed such that type $\tilde{\theta}_i$'s interim utility is the same as the deviation utility from mimicking type $\hat{\theta}_i$. This means the green and black curves intersect at type $\tilde{\theta}_i$. Hence, incentive compatibility in the original mechanism implies that at type $\tilde{\theta}_i$, the green (black) curve is above the red curve, thus also above the blue curve. Given that each agent's utility decreases in the signal (Assumption 1), the green curve above the blue at type $\tilde{\theta}_i$ implies that $\hat{s}_i \leq \Sigma_i(G, \hat{\theta}_i)$. Thus, we establish that our candidate mechanism improves utility for $\hat{\theta}_i$ by removing coordination.

¹⁴Careful readers may wonder why there are multiple crossings between the blue and red curves. Recall that our single-crossing condition only requires comparisons across deterministic outcomes (Assumption 2), whereas a coordination mechanism could offer stochastic recommendations. Thus, the red and blue curves may exhibit multiple crossings (Quah and Strulovici, 2012).

Note that when strict ICE holds, $\Sigma_i(G, \theta_i) < \Sigma_i(G, \hat{\theta}_i)$ for all $\theta_i < \hat{\theta}_i$ if the signal recommendation in G is non-degenerate. This implies that the red curve is strictly above the blue curve at type $\tilde{\theta}_i$. Intuitively, this means that by replacing a stochastic recommendation with a deterministic one, the upward deviation incentive for the lower type $\tilde{\theta}_i$ becomes strictly lower in the deterministic mechanism. This strict slackness in type $\tilde{\theta}_i$'s upward deviation incentive enables us to strictly reduce the deterministic signal recommended to type $\hat{\theta}_i$. In Fig. 1, this means that the green curve is also strictly above the blue curve at type $\tilde{\theta}_i$, and hence $\hat{s}_i < \Sigma_i(G, \hat{\theta}_i)$. To summarize, under the strict ICE, the utility improvement from replacing a stochastic recommendation with a deterministic one is also strict. In other words, coordination mechanisms are strictly sub-optimal under the strict ICE.

Incentive compatibility. Now we argue that (2) holds. Notice that we construct our candidate mechanism so that the local upward-deviation constraint for type $\tilde{\theta}_i$ is binding. Then, given a monotone allocation rule, the single-crossing condition implies that the local downward-deviation constraint for type $\hat{\theta}_i$ also holds.¹⁵ Recall that our single-crossing condition only requires comparisons across deterministic outcomes (Assumption 2); we cannot apply it here to rule out global deviation because our current candidate mechanism may still involve stochastic recommendations for types above $\hat{\theta}_i$. In the formal proof, we circumvent this by modifying the candidate mechanism as follows: construct our candidate mechanism inductively so that the signal recommendations are deterministic for all types, and all local upward-deviation constraints are binding. As a result, the single-crossing condition ensures that all the local downward-deviation constraints also hold. Moreover, the single-crossing property also ensures that no global deviation is profitable.

In our construction, upward-deviation incentives are the primary concern, in contrast to classical revenue-maximizing auction design where downward-deviation incentives typically bind. This distinction stems from the differing design objectives. In revenue maximization, the first-best benchmark (absent incentive constraints) involves charging each agent their full valuation. Private information then introduces downward-deviation incentives, as higher types mimic lower types to retain informational surplus. Conversely, because our objective is to maximize agent utilities, the first-best benchmark for any monotone interim allocation rule would set the signaling cost to zero. Consequently, the economic tension reverses: private information creates upward-deviation incentives, as now lower types want to mimic higher types to secure a larger allocation without the deterrent of high costs. This comparison illustrates why upward-deviation constraints bind in utility-maximization environments, while downward-deviation constraints bind in revenue-maximization settings.

¹⁵Moreover, by construction, both the local upward-deviation constraint and the local downward-deviation constraint for comparisons among other types remain valid in the candidate mechanism.

Payoff non-Equivalence. Theorem 1 operates in an environment where payoff equivalence fails: fixing an implementable (interim) allocation rule does not pin down agents' interim utilities, even after normalizing the lowest type's utility. The main driver for this non-equivalence is the type-dependent signaling cost. As we have seen in our introductory example, different implementations of the same allocation rule can induce different distributions of signal requirements across types. In environments with quasi-linear payments or type-independent signaling costs, different distributions of signal requirements have the same certainty equivalence. However, under type-dependent signaling costs, different distributions of signal requirements result in different certainty equivalences, which eventually translates directly into different information rents.

It is also useful to contrast this source of non-equivalence with the one in signaling games. In signaling games, there is usually one sender (the agent) and one receiver (the principal). There, the same allocation outcome can result in different signaling costs because the principal could have different off-path beliefs. Refinements such as D1 address precisely this belief-driven indeterminacy by selecting the Riley outcome (minimal-cost separating equilibrium). In our model, by contrast, payoff non-equivalence is not driven by belief selection, but reflects real differences in the signaling costs required to implement a given allocation rule. To see this, since there is only one agent in the Spencian signaling game, there is no scope for coordination by assumption: the agent chooses their own signal based solely on their own private type¹⁶. In contrast, in our setting, there is a non-trivial tradeoff in how the designer uses cross-agent information to coordinate signal requirements.

Strict Optimality of Non-coordination. Let us take a closer look at when coordination mechanisms can be strictly outperformed by non-coordination mechanisms. When the costs are type-independent or multiplicatively separable, the certainty equivalence signal is constant over type. In such environments, payoff equivalence holds: once interim allocations are fixed, the details of implementation do not affect agents' utilities. As a result, while there may be many welfare-optimal mechanisms, potentially including coordination, pay-off equivalence ensures that it is without loss of optimality to focus on non-coordination mechanisms. However, when the costs are type-dependent, implementation details matter for welfare. As we have discussed earlier, under the strict ICE condition, any coordinating mechanism can be *strictly* improved by a non-coordination mechanism.

¹⁶An alternative interpretation is that there is a continuum of agents. Regardless of the interpretation, each agent chooses their own signal based solely on their own private type in the Spencian signaling game.

Optimal Non-coordination Mechanism. The optimality of non-coordination substantially reduces the complexity of the design space. By ruling out mechanisms that rely on cross-agent coordination in signal recommendations, it narrows attention to a tractable class of implementations and provides a practical path toward characterization and implementation. For example, in a follow-up work, Li and Qiu (2026) study resource allocation when agents can misrepresent their types and compete with one another. In their setting, each agent has utility $x_i - (s_i - \theta_i)^+$, which satisfies the ICE assumption. As a result, by invoking Theorem 1, they can, without loss of optimality, restrict attention to non-coordination mechanisms, which enables a characterization of the optimal design. Li and Qiu (2026) further apply this characterization in large markets and show that the optimal mechanism may distort the efficient allocation, either at the middle or at the top, in order to maximize agents' utilities.

4 Indirect Implementation: Coarse Ranking Contests

There is a tight connection between non-coordination mechanisms and contests. In this section, we assume that the environment is symmetric, that is, all agents have the same utility function and follow the same type distribution. We will show that in the symmetric environment, coarse ranking contests can implement symmetric direct non-coordination mechanisms, where the allocation rule for each agent is the same. Specifically, we will propose the concept of coarse ranking, in which segments of signals are pooled and assigned the same rank. Correspondingly, we extend the usual notion of a contest to the so-called coarse ranking contest, where the items are allocated to the k agents with the highest coarse rankings, with ties broken uniformly at random.

Definition 3 (coarse ranking). Given any countable set of disjoint open intervals $\{(\underline{s}^{(j)}, \bar{s}^{(j)})\}_{j=1}^{\infty}$ whose union is a subset of the signal space, the *coarse ranking* of agent i under the signal profile $\mathbf{s} = (s_1, \dots, s_n)$ is

$$r_i(\mathbf{s}) = \left| \left\{ i' \neq i, 1 \leq i' \leq n : s_{i'} > \bar{s}^{(j)} \right\} \right|,$$

and the number of ties for agent i is

$$z_i(\mathbf{s}) = \left| \left\{ i' \neq i, 1 \leq i' \leq n : \bar{s}^{(j')} = \bar{s}^{(j)} \right\} \right| + 1.$$

Here, for any signal s_i , j is the index such that $s_i \in (\underline{s}^{(j)}, \bar{s}^{(j)})$, if such a j exists (i.e., if

s_i falls into one of the intervals defined). If no such j exists (i.e., if s_i lies outside all the intervals defined), then (slightly overloading the notation) we let $\bar{s}^{(j)} = \underline{s}^{(j)} = s_i$. We also call the pair of functions $(\mathbf{r}, \mathbf{z})$ a *coarse ranking*.

Intuitively, each interval in the set $\{(\underline{s}^{(j)}, \bar{s}^{(j)})\}_{j=1}^{\infty}$ specifies a region of signals that are pooled and assigned the same (coarse) ranking. Outside the closure of the union of these intervals, signals are ranked strictly. In the special case where $\{(\underline{s}^{(j)}, \bar{s}^{(j)})\}_{j=1}^{\infty}$ is empty, the definition of a coarse ranking coincides with the usual definition of a strict ranking, where the agents' ranks are given by the order of their signals in the given signal profile.

Our generalization of strict rankings to coarse rankings leads to a larger class of contest rules, defined as follows. For any coarse ranking $(\mathbf{r}, \mathbf{z})$ and any agent i , the induced (coarse ranking) contest rule is

$$\tilde{x}_i(\mathbf{s}; \mathbf{r}, \mathbf{z}) = \begin{cases} 1, & k \geq r_i(\mathbf{s}) + z_i(\mathbf{s}), \\ \frac{k - r_i(\mathbf{s})}{z_i(\mathbf{s})}, & k \in (r_i(\mathbf{s}), r_i(\mathbf{s}) + z_i(\mathbf{s})), \\ 0, & k \leq r_i(\mathbf{s}). \end{cases}$$

Definition 4 (coarse ranking contest rule). A mapping profile $\mathbf{x} : \mathbf{S} \rightarrow X$ is a *coarse ranking contest rule* if there exists a coarse ranking $(\mathbf{r}, \mathbf{z})$ such that $x_i(\mathbf{s}) = \tilde{x}_i(\mathbf{s}; \mathbf{r}, \mathbf{z})$ for each i .

The class of coarse ranking contest rules, which is a subset of the class of all mappings from the signal space to the allocation space, also includes the class of contest rules that allocate items (prizes) to agents based on the strict ranking of their signals such as all-pay contests (Baye et al., 1996). The proof of the proposition is provided in Section B.

Proposition 1. *In symmetric environments, any symmetric interim allocation–utility pair (Q, U) that can be implemented by a direct non-coordination mechanism has an indirect implementation that is a randomization over the coarse ranking contests.*

5 Discussions

In this section, we establish a partial converse to our main result, showing that coordination mechanisms can outperform non-coordination mechanisms when agents' utility functions satisfy a strictly decreasing certainty equivalence condition. We show that in our model, it is possible for the principal to implement a non-monotone allocation, and we identify sufficient conditions under which allowing non-monotone allocations does not alter our main conclusions.

5.1 Sub-optimality of Non-coordination

We now present a partial converse to Theorem 1, showing that if Assumption 3 is violated, then direct mechanisms can strictly outperform non-coordination mechanisms. Specifically, when agents' certainty equivalence is strictly decreasing in type, coordination through stochastic signal recommendations can yield strictly higher utility for at least one type, even when the underlying allocation rule remains unchanged.

Assumption 4 (Decreasing certainty equivalence). For any non-degenerate distribution G over allocations and signals and for any agent i , his certainty equivalence $\Sigma_i(G, \theta)$ is strictly decreasing in θ .

Proposition 2 (Sub-optimality of non-coordination). *Suppose Assumption 1, 2, and 4 hold. Fix any $(\mathbf{Q}, \mathbf{U})$ that can be implemented by a non-coordination mechanism and that $\mathbf{Q}$ is strictly increasing. Then there exists a distribution $\mathbf{H}$ and $(\mathbf{Q}^\dagger, \mathbf{U}^\dagger)$ that can be implemented by a direct mechanism with stochastic signal recommendation. Moreover, $\mathbf{Q}^\dagger = \mathbf{Q}$ and $U_i^\dagger(\theta_i) \geq U_i(\theta_i)$ for any i and type θ_i , where the inequality is strict for at least one type.*

Note that Proposition 2 only implies that non-coordination mechanisms can be improved upon by coordination mechanisms with stochastic signal recommendation. The result is silent on which coordination structure is optimal for the implementation. For example, in Section E.1, we introduce a special class of coordination mechanism, the ex post certification mechanism, which serves as an analog of the winner-pays-bid mechanism. Establishing the optimality of the ex post certification mechanism would require further structures on the utility functions of the agents beyond the simple strictly increasing certainty equivalence condition. We leave this as an interesting open question.

5.2 The Implementability of Non-Monotone Allocations

We focus on implementing monotone allocations, motivated by both practical and ethical fairness considerations (see, e.g., Roth, 2002; Gershkov et al., 2022). Monotonicity offers a transparent and easily interpretable structure that aligns with widely held normative intuitions: individuals who demonstrate greater merit, exert more effort, or exhibit greater need should not be disadvantaged relative to others. This principle resonates in domains such as college admissions, where stronger academic records are expected to improve admission chances, and in means-tested programs, where lower incomes should yield more generous benefits (Budish, 2011). In such contexts, monotonicity not only enhances predictability but also reinforces institutional legitimacy by aligning with public expectations and reducing the risk of perceived arbitrariness or political backlash.

In contrast, non-monotone rules are harder to justify in public-facing mechanisms where transparency and accountability are critical. They may conflict with intuitive fairness norms and lead stakeholders to view the process as opaque or capricious. When improved qualifications yield worse outcomes, this can generate confusion, resentment, and erode trust, even when such rules are defensible within a formal model.

Nonetheless, from a theoretical standpoint, it is interesting to explore the boundaries of implementability and consider when non-monotone allocation rules can arise. The following example demonstrates that non-monotone allocations may be implementable.

Example 2. Consider a simplified single-agent setting where the agent’s type θ is drawn from a uniform distribution on $[0, 2]$, and the agent’s utility function is $x - (s - \theta)^+$. Consider the following two options:

The first option has a deterministic signal recommendation: The agent receives an item with probability $\frac{1}{2}$ if the generated signal is $\frac{2}{3}$. The second option has a random signal recommendation: With probability $\frac{1}{2}$, the agent receives an item with probability $\frac{1}{2}$ if the generated signal is $\frac{1}{2}$, and with the remaining probability, the agent receives an item with probability 1 if the generated signal is 2.

It is easy to verify that if the agent has type $\theta \in [\frac{1}{3}, \frac{3}{2}]$, the agent prefers the deterministic signal recommendation, and if the agent has type $\theta \in [0, \frac{1}{3}] \cup [\frac{3}{2}, 2]$, the agent prefers the randomized signal recommendation. This results in an implementable non-monotone allocation.

We can extend this example to a multi-agent setting, showing that when the principal leverages cross-agent information to generate signal recommendations, she can implement a broader class of allocation rules, including some that are non-monotone. However, implementing non-monotone allocations may be suboptimal from the principal’s perspective. In Section 5.3, we demonstrate that under an additional assumption, any implementable non-monotone allocation rule is dominated by a monotone allocation rule, in the sense that implementing the latter via a non-coordination mechanism yields weakly higher utility for all types. Furthermore, when the principal favors assortative matching that assigns higher allocations to higher types, implementing the monotone allocation rule using a non-coordination mechanism strictly improves the principal’s objective.

5.3 Optimality of Monotone Allocations

As illustrated in Example 2, certain non-monotone interim allocations can be implemented via direct mechanisms. This arises when agents’ utility functions fail to satisfy the single-crossing property under stochastic (lottery-based) signal recommendations.

We introduce “no concave crossing”, an assumption that regulates the agent’s preference on deterministic signal recommendation and stochastic signal recommendation. Under this assumption, we can show that any non-monotone allocation rule that can be implemented by a direct mechanism leaves each agent with a lower utility. In other words, we can always find another monotone allocation rule, such that it can be implemented by a direct non-coordination mechanism that leaves each agent with a higher utility. This implies that focusing on monotone allocations in Theorem 1 is without loss of optimality.

Assumption 5 (no-concave-crossing). For any agent i , consider any allocation and deterministic signal recommendation (x_i, s_i) , and any allocation and stochastic signal recommendation (x'_i, G_i) , if there exist two types $\hat{\theta}_i < \hat{\theta}'_i$ such that

$$\begin{aligned} u_i(x_i, s_i, \hat{\theta}_i) &\leq \mathbf{E}_{s \sim G_i} [u_i(x'_i, s, \hat{\theta}_i)] \\ u_i(x_i, s_i, \hat{\theta}'_i) &\leq \mathbf{E}_{s \sim G_i} [u_i(x'_i, s, \hat{\theta}'_i)], \end{aligned}$$

then for any $\theta_i \in [\hat{\theta}_i, \hat{\theta}'_i]$, we have the following $u_i(x_i, s_i, \theta_i) \leq \mathbf{E}_{s \sim G_i} [u_i(x'_i, s, \theta_i)]$.

Note that the above condition compares any deterministic outcome with any lottery. Our condition is weaker than the one studied in Kartik et al. (2024), which considers two arbitrary lotteries. Examples satisfying our “no concave crossing” condition includes $u_i(x_i, s_i, \theta_i) = v_i(x_i, \theta_i) - c_i(s_i, \theta_i)$ when $c_i(s_i, \theta_i) = (s_i - \theta_i)^+$ and $c_i(s_i, \theta_i) = (s_i - \theta_i)^2$.

Theorem 2. *Under Assumptions 5, for any interim allocation–utility pair $(\mathbf{Q}, \mathbf{U})$ that can be implemented by a direct mechanism, there exists another interim allocation–utility pair $(\mathbf{Q}^\dagger, \mathbf{U}^\dagger)$ such that (1) it can be implemented by a non-coordination mechanism; (2) $\mathbf{Q}^\dagger$ is a monotone allocation rule; (3) $U_i^\dagger(\theta_i) \geq U_i(\theta_i)$ for all i .*

The key idea of the proof is to construct a direct non-coordination mechanism that implements the monotone rearrangement of a given (possibly non-monotone) allocation rule, denoted $Q_i^\dagger$. We then show that this rearranged allocation leads to weakly higher utility for all types and strictly higher utility for at least one type. Formally, given any non-monotone allocation rule $\mathbf{Q}$, define its monotone rearrangement $\mathbf{Q}^\dagger$ as follows. For each i , let $G_i(z) = \int_{\bar{\theta}_i}^{\bar{\theta}_i} \mathbf{1}\{Q_i(\theta) \leq z\} dF_i(\theta)$ be the distribution function of the original allocation rule. The **monotone rearrangement** (or quantile transform) of Q_i is defined by:

$$Q_i^\dagger(\theta) = G_i^{-1}(F_i(\theta)),$$

where $G_i^{-1}(u) = \inf\{z \in \mathbb{R} : G_i(z) \geq u\}$ is the generalized inverse of the distribution function G_i .

Recall that the principal’s objective is to maximize a weighted average of agents’ utilities, possibly combined with other goals such as matching efficiency. To compare alternative allocation rules, we consider a reduced-form objective that includes both: the total matching efficiency

$$\sum_{i=1}^n \mathbf{E}_{\theta_i \sim F_i} [W_i(Q_i(\theta_i), \theta_i)]$$

and the weighted average of agents’ utilities. If the principal’s objective favors assortative matching, that is, the total efficiency is higher under the monotone rearrangement $\mathbf{Q}^\dagger$, then implementing $\mathbf{Q}^\dagger$ via a direct non-coordination mechanism strictly improves upon implementing the original non-monotone allocation rule.

6 Conclusions

In this paper, we study the design of optimal screening mechanisms for allocating limited resources among multiple agents based on costly signals. We show that, to minimize signaling costs, it is optimal not to coordinate agents’ signaling choices using others’ reports. That is, non-coordination mechanisms, where each agent’s recommendation depends only on their own type, maximize agents’ utilities while implementing a given allocation rule.

This finding not only identifies a novel welfare gain in screening environments with endogenous costs but also has broader implications for mechanism design. In particular, the optimality of non-coordination significantly reduces the complexity of the design space. By ruling out mechanisms that rely on cross-agent coordination, it simplifies the search for optimal solutions and offers a tractable path forward for practical implementation.

These insights open several promising directions for future research. One is to explore the structure of optimal mechanisms under more general or heterogeneous cost functions, including those that capture non-standard preferences or domain-specific constraints. Another is to study how non-coordination principles extend to dynamic or multi-stage environments, or interact with equity and redistribution goals. Such investigations may yield deeper theoretical characterizations and inform the design of robust, low-cost allocation mechanisms in education, procurement, housing, and beyond.

A Proof of Theorem 1

To simplify the exposition, we assume that the type space is finite. We denote the type space by $\Theta_i = \{\hat{\theta}_i^{(0)}, \dots, \hat{\theta}_i^{(m)}\}$, with $\hat{\theta}_i^{(0)} < \dots < \hat{\theta}_i^{(m)}$. The extension to continuous types can be shown via standard discretization tricks, with details provided in Section A.1.

Fixing any direct mechanism that delivers the interim allocation $\mathbf{Q}$ and the interim utility $\mathbf{U}$, let $G_i^{(k)}$ be the distribution over the allocations and signals recommended to the agent i who reports as type $\hat{\theta}_i^{(k)}$. Let $Q_i^{(k)} \in [0, 1]$ be the probability of allocating the item to agent i induced by $G_i^{(k)}$. Denote the certainty equivalent signal of $G_i^{(k)}$ for type $\hat{\theta}_i^{(k)}$ by

$$s_i^{(k)} \triangleq \Sigma(G_i^{(k)}, \hat{\theta}_i^{(k)}), \quad 0 \leq k \leq m.$$

As illustrated in the main text, the key proof idea is to construct a mechanism that prescribes deterministic signal recommendations for agents and show that this constructed mechanism is implementable by a non-coordination mechanism and improves agents' utilities. The key idea for the construction is to find a deterministic signal recommendation that prevents local upward deviation.

Denote the signal recommended to type $\hat{\theta}_i^{(k)}$ by $\hat{s}_i^{(k)}$, for all $0 \leq k \leq m$. We proceed inductively. Let the signal recommended to type $\hat{\theta}_i^{(0)}$ be $\hat{s}_i^{(0)} := s_i^{(0)}$. For all $k \in \{1, \dots, m\}$ and all $i \in \{1, \dots, n\}$, let $\hat{s}_i^{(k)}$, the signal recommended to type $\hat{\theta}_i^{(k)}$, be the solution of

$$U_i(\hat{\theta}_i^{(k-1)}; Q_i^{(k)}, \hat{s}_i^{(k)}) = U_i(\hat{\theta}_i^{(k-1)}; Q_i^{(k-1)}, \hat{s}_i^{(k-1)}). \quad (\text{construction})$$

Let $U_i^\dagger(\hat{\theta}_i^{(k)}) := U_i(\hat{\theta}_i^{(k)}; Q_i^{(k)}, \hat{s}_i^{(k)})$. By construction, we have $U_i^\dagger(\hat{\theta}_i^{(0)}) = U_i(\hat{\theta}_i^{(0)})$.

It remains to show that (1) $(\mathbf{Q}, \mathbf{U}^\dagger)$ can be implemented as a direct non-coordination mechanism (**Step 1**); and (2) $U_i^\dagger(\hat{\theta}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)})$ for all i and k (**Step 2**).

Step 1: Recall that, as defined in Definition 1, to show implementability, we need to show consistency and incentive compatibility. First, we show consistency. Given that $\mathbf{Q}$ is an interim allocation, there exists some ex post allocation rule $\mathbf{y} : \Theta \rightarrow X$ that induces $\mathbf{Q}$. If the signal recommendation $\tilde{s}_i(\theta_i)$ is an injection, we can perfectly infer the type from the recommended signals, and by letting $x_i(\mathbf{s}) := y_i(\theta_1(s_1), \dots, \theta_n(s_n))$ where $\theta_i(s_i)$ is the type that recommends signal s_i , we implement the interim allocation $\mathbf{Q}$ under the non-coordination mechanism. If there are multiple types that are recommended with the same signal, by the incentive compatibility constraints, their interim allocations must be the same. Therefore, we can simulate $\theta_i(s_i)$ according to the prior distribution conditional on the recommended signal being s_i , and let $x_i(\mathbf{s}) := y_i(\theta_1(s_1), \dots, \theta_n(s_n))$. This again ensures the implementation of the interim allocation $\mathbf{Q}$. Second, we show that incentive compatibility holds. By construction, there is no incentive for local upward deviation. Given that each agent's utility is weakly increasing in type (Assumption 1) and the fact that local upward deviation incentives are binding, the weak single crossing property (Assumption 2) implies that there is no incentive for local downward deviation. Applying the single crossing property again,

local incentive compatibility ensures global incentive compatibility. Therefore, $(\mathbf{Q}, \mathbf{U}^\dagger)$ can be implemented as a direct non-coordination mechanism.

Step 2: To show improvement in utility, the key is to show that the deterministic signal recommendation for each type in the constructed mechanism is lower than the certainty equivalent signal given the stochastic recommendation in the original mechanism. We prove by induction. Fix any i , suppose the following induction hypothesis holds.

$$U_i^\dagger(\hat{\theta}_i^{(k-1)}) \geq U_i(\hat{\theta}_i^{(k-1)}) \quad (\text{induction})$$

Claim 1. Since $(\mathbf{Q}, \mathbf{U})$ can be implemented as a direct mechanism, we must have the following incentive compatibility condition holds.

$$U_i(\hat{\theta}_i^{(k-1)}) \geq U_i(\hat{\theta}_i^{(k-1)}; Q_i^{(k)}, s_i^{(k)}). \quad (\text{incentive})$$

Let us take a detour and understand the remaining step intuitively using Fig. 1. Suppose $\hat{\theta}_i^{(k-1)}$ is the type $\tilde{\theta}_i$ and $\hat{\theta}_i^{(k)}$ is the type $\hat{\theta}_i$ in Fig. 1. Intuitively, (incentive) guarantees that the red curve is above the blue curve at $\tilde{\theta}_i$. Notice that (construction) ensures that each type is indifferent between local upward deviation and truthful reporting in the constructed candidate mechanism. Intuitively, (construction) means that the value of the green curve at $\tilde{\theta}_i$ equals type $\tilde{\theta}_i$'s expected utility from truthful reporting. Together with (induction), we can infer that the green curve is above the red curve at type $\tilde{\theta}_i$. Hence, the green curve is above the blue curve at type $\tilde{\theta}_i$. This implies that $\hat{s}_i^{(k)} \leq s_i^{(k)}$.

Formally, combining (construction), (induction), and (incentive), we have

$$U_i(\hat{\theta}_i^{(k-1)}; Q_i^{(k)}, \hat{s}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k-1)}; Q_i^{(k)}, s_i^{(k)}).$$

This implies that $\hat{s}_i^{(k)} \leq s_i^{(k)}$. In other words, the deterministic signal recommendation for each type in the constructed mechanism is lower than the certainty equivalent signal. As a result, this improves each type's utility in the constructed mechanism: $U_i^\dagger(\hat{\theta}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)})$.

Induction implies that $U_i^\dagger(\hat{\theta}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)})$ for all i and k , and combining it with Step 1, Theorem 1 holds.

Proof of Claim 1. We use $H_i^{(k)}$ to denote the marginal distribution over allocation given to agent i who reports type $\hat{\theta}_i^{(k)}$. Notice that the mean of $H_i^{(k)}$ is $Q_i^{(k)}$. Denote the certainty equivalent signal of $G_i^{(k)}$ for type $\hat{\theta}_i^{(j)}$ by

$$s_i^{(k,j)} \triangleq \Sigma(G_i^{(k)}, \hat{\theta}_i^{(j)}), \quad 0 \leq k, j \leq m.$$

Moreover, let $s_i^{(k)} := s_i^{(k,k)}$, for all $0 \leq k \leq m$.

First we replace the randomized signal recommendation with its certainty equivalence $s_i^{(k)}$ for each type $\hat{\theta}_i^{(k)}$ while maintaining the same allocation rule. We next show that when type $\hat{\theta}_i^{(k-1)}$ misreports as the adjacent type $\hat{\theta}_i^{(k)}$, his expected utility is weakly lower after the replacement than the one before the replacement.

Type $\hat{\theta}_i^{(k-1)}$'s utility for misreporting as type $\hat{\theta}_i^{(k)}$ in the direct mechanism is at least¹⁷

$$U_i \left(\hat{\theta}_i^{(k-1)}; G_i^{(k)} \right) = U_i \left(\hat{\theta}_i^{(k-1)}; H_i^{(k)}, s_i^{(k,k-1)} \right) \geq U_i \left(\hat{\theta}_i^{(k-1)}; H_i^{(k)}, s_i^{(k)} \right), \quad (4)$$

where the equality follows from the definition of certainty equivalent signal, and the inequality holds since (1) $s_i^{(k,j)}$ is increasing in j by Assumption 3; and (2) the utility function is weakly decreasing in the signal. By incentive compatibility in the original mechanism, we have

$$U_i \left(\hat{\theta}_i^{(k-1)} \right) := U_i \left(\hat{\theta}_i^{(k-1)}; G_i^{(k-1)} \right) \geq U_i \left(\hat{\theta}_i^{(k-1)}; G_i^{(k)} \right). \quad (5)$$

Combining (4) and (5), we conclude that

$$U_i \left(\hat{\theta}_i^{(k-1)} \right) \geq U_i \left(\hat{\theta}_i^{(k-1)}; H_i^{(k)}, s_i^{(k)} \right). \quad (6)$$

That is, type $\hat{\theta}_i^{(k-1)}$ has no incentive to deviate to the higher adjacent type after we replace the signal recommendation with the certainty equivalent signal of the corresponding stochastic recommendation in the original mechanism. Because each agent's utility admits an expected utility representation, we have $U_i(\theta; G_X, s) := Q \cdot u_i(1, s, \theta) + (1-Q) \cdot u_i(0, s, \theta) = U_i(\theta; Q, s)$, where Q is the mean of G_X . Hence (6) implies (incentive). $\square$

A.1 Continuous Types

In this section, we prove Theorem 1 under continuous type space. We can utilize our constructive proof in the finite type space to construct a superior candidate non-coordination mechanism in the continuous type space. The key challenge is to construct the signal recommendation function, since the inductive argument no longer works in the continuous type space. To circumvent this, our idea is to use step functions to approximate the signal recommendation function (**Step 1 & 2**). When we discretize the type space, we can construct the signal recommendation function on the discretized grids as in the proof of Theorem 1

¹⁷Notice that the agent can potentially adopt double deviation strategies in the direct mechanism based on the signal realizations, i.e., after misreporting, the agent can walk away based on the signal realization.

(Step 1). For every type that is not on the discretized grids, we can assign the signal recommendation of the rightmost grid smaller than this type as its signal recommendation. This way, we can construct a sequence of increasing step functions as an approximation to the signal recommendation function in the continuous type space, which admits a converging subsequence **(Step 2)**. Correspondingly, we can also obtain the limit of the utility functions in this construction. This limit gives us a candidate mechanism. It remains to check whether incentive compatibility holds in the limit. Given the smoothness in the utility functions, we can ensure that IC holds in the limit **(Step 3)**. This completes our construction of a candidate mechanism, the limit of the discretized approximation, which is an incentive-compatible non-coordination mechanism that outperforms any given non-coordination one.

Now, let us begin our formal proof. Since both $Q_i(\cdot)$ and $U_i(\cdot)$ are weakly increasing, each has at most countably many discontinuity points. We first define the discretizations based on the discontinuity points.

For each $m \geq 1$, choose a finite set $\Theta_i^{(m)} = \{\hat{\theta}_i^{(m,0)} < \hat{\theta}_i^{(m,1)} < \dots < \hat{\theta}_i^{(m,K_m)}\} \subseteq \Theta_i$ such that $\Theta_i^{(m)} \subseteq \Theta_i^{(m+1)}$, and for any m ,

1. $\max_{k=1,\dots,K_m} (\hat{\theta}_i^{(m,k)} - \hat{\theta}_i^{(m,k-1)}) \leq \frac{1}{m}$;
2. every discontinuity point of $Q_i(\cdot)$ and $U_i(\cdot)$ belongs to $\Theta_i^{(m)}$ for all sufficiently large m (e.g. enumerate discontinuities and ensure the first m are in $\Theta_i^{(m)}$).

Define the left-rounding map $r_i^{(m)} : \Theta_i \rightarrow \Theta_i^{(m)}$ by

$$r_i^{(m)}(\theta_i) := \max\{\hat{\theta}_i \in \Theta_i^{(m)} : \hat{\theta}_i \leq \theta_i\}.$$

Then $r_i^{(m)}(\theta_i) \rightarrow \theta_i$ as $m \rightarrow \infty$ and, by construction of $\Theta_i^{(m)}$, as $m \rightarrow \infty$, we have for every $\theta_i \in \Theta_i$:

$$Q_i(r_i^{(m)}(\theta_i)) \rightarrow Q_i(\theta_i), \quad U_i(r_i^{(m)}(\theta_i)) \rightarrow U_i(\theta_i). \quad (7)$$

Step 1: Constructing a deterministic mechanism on the discretized grids.

Restrict (Q_i, U_i) to the finite set $\Theta_i^{(m)}$. Apply the finite-type version of Theorem 1 to construct an alternative mechanism $(Q_i, U_i^\dagger)$. Hence, we obtain a deterministic signal recommendation function

$$s_i^{(m)} : \Theta_i^{(m)} \rightarrow \mathbb{R}_+ \quad (\text{weakly increasing in } \hat{\theta}_i)$$

such that for all $\hat{\theta}_i, \hat{\theta}'_i \in \Theta_i^{(m)}$,

$$U_i^\dagger(Q_i(\hat{\theta}_i), s_i^{(m)}(\hat{\theta}_i), \hat{\theta}_i) \geq U_i(\hat{\theta}_i), \quad (8)$$

$$U_i^\dagger(Q_i(\hat{\theta}_i), s_i^{(m)}(\hat{\theta}_i), \hat{\theta}_i) \geq U_i^\dagger(Q_i(\hat{\theta}'_i), s_i^{(m)}(\hat{\theta}'_i), \hat{\theta}_i). \quad (9)$$

Step 2: Extending the construction to the whole type space. Extend $s_i^{(m)}$ to all Θ_i by

$$s_i^{(m)}(\theta_i) := s_i^{(m)}(r_i^{(m)}(\theta_i)),$$

which is weakly increasing in θ_i . To extract a pointwise convergent subsequence in $\{s_i^{(m)}(\theta_i)\}_m$, consider a bounded monotone transform $\phi(s) := \frac{s}{1+s} \in [0, 1]$. Then $\phi \circ s_i^{(m)}$ is weakly increasing and bounded in $[0, 1]$. By Helly's selection theorem, there exists a subsequence (still indexed by m) and a weakly increasing function $\sigma_i : \Theta_i \rightarrow [0, 1]$ such that as $m \rightarrow \infty$,

$$\phi(s_i^{(m)}(\theta_i)) \rightarrow \sigma_i(\theta_i) \quad \forall \theta_i \in \Theta_i.$$

Define the pointwise limit of the signal recommendation as

$$s_i(\theta_i) := \phi^{-1}(\sigma_i(\theta_i)) = \frac{\sigma_i(\theta_i)}{1 - \sigma_i(\theta_i)}.$$

Under our maintained model assumption that certainty equivalents exist as finite signals, we may take the subsequence so that $\sigma_i(\theta_i) < 1$ for all θ_i ; hence $s_i(\theta_i) \in \mathbb{R}_+$.

Step 3: Verifying IC and utility dominance in the limit. Fix the converging subsequence identified in Step 2 and relabel it as m . Fix $\theta_i, \theta'_i \in \Theta_i$ and set $\hat{\theta}_{i,m} := r_i^{(m)}(\theta_i)$ and $\hat{\theta}'_{i,m} := r_i^{(m)}(\theta'_i)$. Apply (9) with $(\hat{\theta}_i, \hat{\theta}'_i) = (\hat{\theta}_{i,m}, \hat{\theta}'_{i,m})$:

$$U_i^\dagger(Q_i(\hat{\theta}_{i,m}), s_i^{(m)}(\hat{\theta}_{i,m}), \hat{\theta}_{i,m}) \geq U_i^\dagger(Q_i(\hat{\theta}'_{i,m}), s_i^{(m)}(\hat{\theta}'_{i,m}), \hat{\theta}_{i,m}).$$

Take $m \rightarrow \infty$. Using $\hat{\theta}_{i,m} \rightarrow \theta_i$, $\hat{\theta}'_{i,m} \rightarrow \theta'_i$, the convergence in (7), the observation that $s_i^{(m)}(\hat{\theta}_{i,m}) \rightarrow s_i(\theta_i)$, $s_i^{(m)}(\hat{\theta}'_{i,m}) \rightarrow s_i(\theta'_i)$, and the continuity of the utility function, we obtain

$$U_i^\dagger(Q_i(\theta_i), s_i(\theta_i), \theta_i) \geq U_i^\dagger(Q_i(\theta'_i), s_i(\theta'_i), \theta_i) \quad \forall \theta_i, \theta'_i \in \Theta_i. \quad (10)$$

Thus, the mechanism $\theta_i \mapsto (Q_i(\theta_i), s_i(\theta_i))$ is incentive compatible.

Similarly, apply (8) at $\hat{\theta}_i = \hat{\theta}_{i,m}$ and take $m \rightarrow \infty$ to get

$$U_i^\dagger(Q_i(\theta_i), s_i(\theta_i), \theta_i) \geq U_i(\theta_i) \quad \forall \theta_i \in \Theta_i. \quad (11)$$

Equations (10)–(11) complete the continuous-type extension.

B Proof of Proposition 1

We first recap the definition and the characterizations of feasible interim allocation rules from Border (1991) and Che et al. (2013).

Definition 5 (feasibility). Given any interim allocation rule $\mathbf{Q} = (Q_i)_{i=1}^n$ where $Q_i : \Theta_i \rightarrow [0, 1]$, $\mathbf{Q}$ is *feasible* if there exists an ex-post allocation rule $\mathbf{q}$ such that $Q_i(\theta_i) = \mathbf{E}_{\theta_{-i}}[q_i(\theta_i, \theta_{-i})]$ for any agent i and type θ_i . We say an interim allocation–utility pair $(\mathbf{Q}, \mathbf{U})$ is *feasible* if $\mathbf{Q}$ is feasible.

Lemma 1 (Che et al., 2013). *Given a set $\mathbf{A} = \prod_{i=1}^n A_i \subset \Theta$, let $w(\boldsymbol{\theta}, \mathbf{A}) = |\{i : \theta_i \in A_i\}|$ be the number of agents whose type θ_i is in A_i .¹⁸ The interim allocation rule $\mathbf{Q}$ is interim feasible if and only if*

$$\sum_i \int_{A_i} Q_i(\theta_i) dF_i(\theta_i) \leq \int_{\mathbf{A}} \min\{k, w(\boldsymbol{\theta}, \mathbf{A})\} d\mathbf{F}(\boldsymbol{\theta}) \quad \forall \mathbf{A} = \prod_{i=1}^n A_i \subset \Theta. \quad (\text{IF})$$

Moreover, for monotone allocations in symmetric environments, (IF) is equivalent to the following.¹⁹

$$\int_{\theta}^{\bar{\theta}} Q(z) dF(z) \leq \int_{\theta}^{\bar{\theta}} Q_E(z) dF(z), \quad \forall \theta \in [\underline{\theta}, \bar{\theta}], \quad (\widehat{\text{IF}})$$

where $Q_E(\theta) = \sum_{j=0}^{k-1} \binom{n-1}{j} \cdot (1-F(\theta))^j \cdot F^{n-1-j}(\theta)$ is the interim allocation rule for allocating k items efficiently.

Proof of Proposition 1. For any symmetric interim allocation–utility pair (Q, U) that is implementable by a non-coordination mechanism, by definition, there exists a mapping from the signal space to the allocation space $x(\hat{s})$ specifying the allocation for each agent given the generated signal $\hat{s}$.²⁰ The distribution over types $F(\theta)$ induces a distribution over signals; call it $\hat{F}(s)$. Similarly, a feasibility constraint on the allocation rule defined in the signal space may be induced by $(\widehat{\text{IF}})$ based on $\hat{F}$. Such operations are valid since the recommended signal is a non-decreasing function of the type. By Theorems 1 and 2 in Kleiner

¹⁸Here, $|\cdot|$ denotes the cardinality of a set.

¹⁹In a symmetric environment, by a slight abuse of notation, we use $F = F_i$ for all i to denote each agent's type distribution.

²⁰Notice that such a mapping might not exist if (Q, U) is implementable by a direct mechanism that is not a non-coordination mechanism.

et al. (2021), any monotone feasible allocation rule x can be written as a convex combination of the extreme points. Using the construction of the extreme points in Theorem 3 of Kleiner et al. (2021), one can easily verify that the extreme points are the coarse ranking contest rules defined in Definition 4. Hence x can be expressed as a randomization over the coarse ranking contest rules. Notice that the above operations do not affect the agents' incentives by preserving the distribution over outcomes; hence the expected utility of each agent in the coarse ranking contest is still U . $\square$

C Proof of Proposition 2

Given any utility function of the agents, consider a type distribution where H_i has binary support for all agent i , denoted by $\{\underline{\theta}_i, \bar{\theta}_i\}$ where $\underline{\theta}_i < \bar{\theta}_i$. Given any strictly increasing interim allocation Q_i , let $\underline{s}_i$ ($\bar{s}_i$) be the signals recommended to type $\underline{\theta}_i$ (type $\bar{\theta}_i$) in the non-coordination mechanism. It is easy to verify that in the optimal non-coordination mechanism, $\underline{s}_i = 0$ and $\underline{s}_i < \bar{s}_i$.

Since the utility function is continuous and strictly decreasing in s and $\bar{s}_i$ is strictly positive, there exists a non-degenerate distribution G_i over signal recommendations such that an agent with type $\bar{\theta}_i$ is indifferent between signal recommendation G_i and deterministic signal recommendation $\bar{s}_i$.²¹ Moreover, by Assumption 4, the utility of type $\underline{\theta}_i$ for misreporting as type $\bar{\theta}_i$ is strictly lower given G_i compared to deterministic signal recommendation $\bar{s}_i$. Therefore, there exists G'_i that first-order stochastically dominates G_i such that the utility of type $\underline{\theta}_i$ for obtaining signal recommendation G'_i and $\bar{s}_i$ are the same. By offering signal recommendation $\underline{s}_i$ to type $\underline{\theta}_i$ with interim allocation $Q_i(\underline{\theta}_i)$, and offering randomized signal recommendation G'_i to type $\bar{\theta}_i$ with interim allocation $Q_i(\bar{\theta}_i)$ in the general mechanism, the utility of all types weakly improves, and the utility of type $\bar{\theta}_i$ strictly improves.

D Monotone Allocations

To simplify the exposition, we assume that the type space of agent i is discrete and finite; that is, $\Theta_i = \{\hat{\theta}_i^{(0)}, \dots, \hat{\theta}_i^{(m)}\}$, where $\hat{\theta}_i^{(0)} < \dots < \hat{\theta}_i^{(m)}$.

Let $\mathbf{Q}^\dagger$ be a monotone rearrangement of $\mathbf{Q}$ that is feasible. We will construct a $\mathbf{U}^\dagger$ such that $(\mathbf{Q}^\dagger, \mathbf{U}^\dagger)$ is implementable by a non-coordination mechanism and show that $U_i^\dagger(\hat{\theta}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)})$ for all i and all $k \in \{0, \dots, m\}$.

²¹Such a non-degenerate distribution G_i exists since $\bar{s}_i > 0$. We can construct it via a distribution with binary support $\bar{s}_i - \epsilon$ and $\bar{s}_i + \epsilon$, where $\epsilon \in (0, \bar{s}_i)$ and the probability of $\bar{s}_i + \epsilon$ is q_ϵ . The strict monotonicity of utility in costly signals implies that q_ϵ must be an interior solution, i.e., $q_\epsilon \in (0, 1)$.

When $k = 0$, let $\hat{s}_i^{(0)} := 0$. Let $U_i^\dagger(\hat{\theta}_i^{(0)}) = U_i(\hat{\theta}_i^{(0)}, Q_i^\dagger(\hat{\theta}_i^{(0)}), \hat{s}_i^{(0)})$. Since $\mathbf{Q}^\dagger$ is a monotone rearrangement of $\mathbf{Q}$, we have $Q_i^\dagger(\hat{\theta}_i^{(0)}) \geq \min_{\hat{\theta}_i \in \Theta_i} Q_i(\hat{\theta}_i)$. Therefore,

$$U_i^\dagger(\hat{\theta}_i^{(0)}) \geq \min_{\hat{\theta}_i \in \Theta_i} U_i(\hat{\theta}_i, Q_i^\dagger(\hat{\theta}_i^{(0)}), 0) \geq \min_{\hat{\theta}_i \in \Theta_i} U_i(\hat{\theta}_i) \geq U_i(\hat{\theta}_i^{(0)})$$

where the last inequality holds since the interim utility must be monotone increasing in the type of the agent and $\hat{\theta}_i^{(0)}$ is the lowest type.

Now consider any $k \geq 1$. Suppose $Q_i^\dagger(\hat{\theta}_i^{(k)}) \geq Q_i(\hat{\theta}_i^{(k)})$. We can apply the same argument as in Theorem 1 to construct $U_i^\dagger(\hat{\theta}_i^{(k)})$.

Suppose alternatively $Q_i^\dagger(\hat{\theta}_i^{(k)}) < Q_i(\hat{\theta}_i^{(k)})$. Let $\bar{s}_i^{(k)}$ be the signal such that type $\hat{\theta}_i^{(k-1)}$ is indifferent between truthfully reporting and misreporting as type $\hat{\theta}_i^{(k)}$:

$$U_i^\dagger(\hat{\theta}_i^{(k-1)}) = U_i(\hat{\theta}_i^{(k-1)}; Q_i^\dagger(\hat{\theta}_i^{(k)}), \bar{s}_i^{(k)}).$$

Let $\bar{s}_i^{(k)}$ be the signal recommended for typing $\hat{\theta}_i^{(k)}$. Hence $U_i^\dagger(\hat{\theta}_i^{(k)}) = U_i(\hat{\theta}_i^{(k)}; Q_i^\dagger(\hat{\theta}_i^{(k)}), \bar{s}_i^{(k)})$.

Following similar arguments as in Theorem 1, we can show that $(\mathbf{Q}^\dagger, \mathbf{U}^\dagger)$ can be implemented by a direct non-coordination mechanism. It remains to show that $U_i^\dagger(\hat{\theta}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)})$.

Since $\mathbf{Q}^\dagger$ is a monotone rearrangement of $\mathbf{Q}$, for any i and any type $\hat{\theta}_i^{(k)}$, there must exist $k' > k$ such that $Q_i(\hat{\theta}_i^{(k')}) \leq Q_i^\dagger(\hat{\theta}_i^{(k)})$. Let $\tilde{s}_i$ be the signal that type $\hat{\theta}_i^{(k-1)}$ is indifferent between truthfully reporting in the direct non-coordination mechanism and consuming $(Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i)$:

$$U_i^\dagger(\hat{\theta}_i^{(k-1)}) = U_i(\hat{\theta}_i^{(k-1)}; Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i). \quad (12)$$

Since type $\hat{\theta}_i^{(k-1)}$ is indifferent between these two options, $Q_i^\dagger(\hat{\theta}_i^{(k)}) \geq Q_i(\hat{\theta}_i^{(k')})$ implies that $\bar{s}_i^{(k)} \geq \tilde{s}_i$. Moreover, by Assumption 2 (wSCP), we have that for type $\hat{\theta}_i^{(k)}$,

$$U_i(\hat{\theta}_i^{(k)}; Q_i^\dagger(\hat{\theta}_i^{(k)}), \bar{s}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)}; Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i). \quad (13)$$

Let $G_i^{(k)}$ be the distribution over signal recommendations to type $\hat{\theta}_i^{(k)}$ under the original mechanism that gives agent i interim utility U_i . We first establish the following two inequalities that describe the preference of type $\hat{\theta}_i^{(k')}$ and type $\hat{\theta}_i^{(k-1)}$.

Claim 2. Type $\hat{\theta}_i^{(k')}$ prefers receiving the deterministic recommendation $(Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i)$ to $(Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)})$ in the original direct mechanism:

$$U_i(\hat{\theta}_i^{(k')}; Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i) \geq U_i(\hat{\theta}_i^{(k')}; Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)}).$$

Claim 3. Type $\hat{\theta}_i^{(k-1)}$ prefers receiving the deterministic recommendation $(Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i)$ to $(Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)})$ in the original direct mechanism:

$$U_i(\hat{\theta}_i^{(k-1)}; Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i) \geq U_i(\hat{\theta}_i^{(k-1)}; Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)}).$$

Since $\hat{\theta}_i^{(k-1)} < \hat{\theta}_i^{(k)} < \hat{\theta}_i^{(k')}$, by Assumption 5, we know that type $\hat{\theta}_i^{(k)}$ also prefers receiving the deterministic recommendation $(Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i)$ to $(Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)})$ in the original direct mechanism:

$$U_i(\hat{\theta}_i^{(k)}; Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i) \geq U_i(\hat{\theta}_i^{(k)}; Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)}). \quad (14)$$

(13) and (14) imply that $U_i(\hat{\theta}_i^{(k)}; Q_i^\dagger(\hat{\theta}_i^{(k)}), \bar{s}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)}; Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)})$. Notice that by construction, the left-hand side is $U_i^\dagger(\hat{\theta}_i^{(k)})$ and by definition, the right-hand side is $U_i(\hat{\theta}_i^{(k)})$. Hence, we can conclude that $U_i^\dagger(\hat{\theta}_i^{(k)}) \geq U_i(\hat{\theta}_i^{(k)})$.

Proof of Claim 2. By the inductive argument $U_i^\dagger(\hat{\theta}_i^{(k-1)}) \geq U_i(\hat{\theta}_i^{(k-1)})$. By (12), the definition of $\tilde{s}_i$, we have $\tilde{s}_i \leq s_i^{(k',k)}$. Moreover, we already know that by Assumption 3, $s_i^{(k',k)} \leq s_i^{(k')}$. These together imply that

$$U_i(\hat{\theta}_i^{(k')}; Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i) \geq U_i(\hat{\theta}_i^{(k')}), \quad (15)$$

By incentive compatibility in the original direct mechanism, we have

$$U_i(\hat{\theta}_i^{(k')}) \geq U_i(\hat{\theta}_i^{(k')}; Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)}). \quad (16)$$

Combining the two, we have the desired inequality. $\square$

Proof of Claim 3. This is because

$$U_i(\hat{\theta}_i^{(k-1)}; Q_i(\hat{\theta}_i^{(k')}), \tilde{s}_i) = U_i^\dagger(\hat{\theta}_i^{(k-1)}) \geq U_i(\hat{\theta}_i^{(k-1)}) \geq U_i(\hat{\theta}_i^{(k-1)}; Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)}).$$

The first inequality holds by the induction assumption. The second inequality holds because $U_i(\hat{\theta}_i^{(k-1)}; Q_i(\hat{\theta}_i^{(k)}), G_i^{(k)})$ is the utility that type $\hat{\theta}_i^{(k-1)}$ obtains by deviating to report type $\hat{\theta}_i^{(k)}$ and always following the signal recommendation. $\square$

E Non-separable Utilities

In this section, we clarify why Assumption 3 may fail under most non-separable preferences. Then we propose a weaker certainty-equivalence monotonicity requirement. We show that under this weaker requirement, we can extend the comparison between any non-coordination

implementation and a simple form of coordination implementations. A similar analysis of ranking the performance of simple mechanisms has also been investigated in the auction context under different objectives and utility models (e.g., Milgrom and Weber, 1982; Matthews, 1987).

Recall that Assumption 3 requires that for *every* distribution G over allocations and recommended signals, the certainty-equivalent signal $\Sigma(G, \theta)$ is increasing in θ . This can fail even for commonly used non-separable expected-utility representations. For example, under the preference that exhibits constant absolute risk aversion: $u_i(x_i, s_i, \theta_i) = 1 - \exp(-a(\theta_i \cdot x_i - s_i))$ for some constant $a > 0$, Assumption 3 is violated.

However, we highlight that for some design questions, one does not need ICE for *all* lotteries. Below, we propose a weaker monotonicity property that is sufficient to compare *non-coordination* (all-pay type formats) and *ex post certification* (winner-only formats).

E.1 Ex Post Certification Mechanisms

We formalize a subclass of direct mechanisms where *only tentative winners* are ever asked to produce a positive signal.

Definition 6 (Ex post certification mechanism). A direct mechanism $(\tilde{\mathbf{s}}, \mathbf{y})$ is an *ex post certification mechanism* if there exist a *tentative winner selection rule* $\hat{y} : \Theta \rightarrow X$ and a *certification schedule* $\hat{s} : \Theta_i \rightarrow \mathbb{R}_+$ such that the following holds.

1. For every report profile $\hat{\theta} \in \Theta$, the mechanism draws a tentative winner set W according to $\hat{y}(\hat{\theta})$.
2. Given $(\hat{\theta}, W)$, the recommendation to agent i is

$$\tilde{s}_i(\hat{\theta}, W) = \begin{cases} \hat{s}(\hat{\theta}_i), & \text{if } i \in W, \\ 0, & \text{if } i \notin W. \end{cases}$$

Thus, each agent is asked to certify only when he is selected as a tentative winner.

3. The allocation rule awards the item to i if and only if i is a tentative winner and he follows the recommendation, i.e., $y_i(\hat{\theta}, W, s') = \mathbf{1}\{i \in W\} \cdot \mathbf{1}\{s'_i = \hat{s}(\hat{\theta}_i)\}$.

The above representation captures the idea that the designer first selects tentative winners (potentially using cross-agent information), and then requires only tentative winners to produce costly certificates. This is the natural analog of “winner-pays” formats in standard auction theory, whereas non-coordination mechanisms correspond to “all-pay” formats.

E.2 A Weaker Monotonicity Condition

Note that in any ex post certification mechanism, each agent faces a lottery over signals. This lottery has binary support, and one potential signal is 0. Therefore, to show the superiority of non-coordination mechanisms over ex post certification mechanisms, it suffices to consider binary lotteries before converting them to certainty equivalences. This motivates our following definition.

Assumption 6 (Winner-only Increasing Certainty Equivalence (WICE)). A utility function u_i is said to have (strict) winner-only increasing certainty equivalence if for any distribution G over allocations and signals with binary support on $(0, 0)$ and $(1, s)$ for some $s \in \mathbb{R}_+$, his certainty equivalent signal $\Sigma_i(G, \theta)$ is (strictly) increasing in θ .

Assumption 6 is strictly weaker than ICE (Assumption 3) because it only considers *binary* lotteries rather than any arbitrary distribution over allocations and signals. To better understand Assumption 6, we provide a simple sufficient condition on the primitives such that Assumption 6 holds. We emphasize that this condition is *not* meant to be necessary for all non-separable utility functions; rather, it is a transparent sufficient condition tailored to the ex post certification environment to facilitate our understanding.

Lemma 2 (Sufficient Condition for WICE). *Suppose Assumption 1 holds. In addition, assume:*

1. *there exists a weakly decreasing function $u^0(s)$ such that $u(0, s, \theta) = u^0(s)$ for all (s, θ) ;*
2. *the utility function satisfies decreasing differences in (θ, s) when $x = 1$, i.e., for any $\theta' > \theta$ and any $s' > s$,*

$$u(1, s', \theta') - u(1, s, \theta') \leq u(1, s', \theta) - u(1, s, \theta). \quad (17)$$

Then the utility function satisfies Assumption 6 (WICE).

The first condition in Lemma 2 states that when an agent does *not* receive the resource ($x = 0$), his utility from producing a costly certificate s is purely a type-independent “wasted effort” term. This is realistic in applications where the type θ captures the valuation of the resource, because the type is only relevant for utility if the agent receives the resource. The second condition in Lemma 2 is a standard *submodularity* (or decreasing-differences) condition. Economically, it states that a higher certificate requirement compresses payoff differences across types in the winning state.²² Put differently, a higher certificate reduces the incremental utility of having a higher type.

²²If u is C^2 in (s, θ) , then (17) is equivalent to $u_{\theta s}(1, s, \theta) \leq 0$ for all (s, θ) .

Proof of Lemma 2. Fix any binary distribution G with support $\{(0, 0), (1, \bar{s})\}$ for some $\bar{s} \geq 0$, and let $q := \Pr_G(x = 1) \in [0, 1]$. If $q \in \{0, 1\}$ or $\bar{s} = 0$, then $\Sigma(G, \theta)$ is constant in θ and the claim is immediate. Hence assume $q \in (0, 1)$ and $\bar{s} > 0$.

Let $c(\theta) := \Sigma(G, \theta)$. By definition, $c(\theta)$ satisfies

$$(1 - q) u^0(c(\theta)) + q u(1, c(\theta), \theta) = (1 - q) u^0(0) + q u(1, \bar{s}, \theta). \quad (18)$$

Rearranging (18) gives

$$q \left(u(1, c(\theta), \theta) - u(1, \bar{s}, \theta) \right) = (1 - q) \left(u^0(0) - u^0(c(\theta)) \right) \geq 0, \quad (19)$$

where the inequality uses that $u^0(\cdot)$ is (weakly) decreasing. Since $u(1, \cdot, \theta)$ is (weakly) decreasing, (19) implies $c(\theta) \leq \bar{s}$.

Now fix $\theta' > \theta$ and set $c := c(\theta)$. Consider the certainty-equivalence equation for type θ' evaluated at c :

$$\begin{aligned} & (1 - q) u^0(c) + q u(1, c, \theta') - (1 - q) u^0(0) - q u(1, \bar{s}, \theta') \\ &= \underbrace{\left((1 - q) u^0(c) + q u(1, c, \theta) - (1 - q) u^0(0) - q u(1, \bar{s}, \theta) \right)}_{=0 \text{ by (18)}} \\ &+ q \left([u(1, c, \theta') - u(1, c, \theta)] - [u(1, \bar{s}, \theta') - u(1, \bar{s}, \theta)] \right). \end{aligned}$$

By the decreasing-differences condition in the winner state, the type premium $u(1, s, \theta') - u(1, s, \theta)$ is weakly decreasing in s . Since $c \leq \bar{s}$, we have

$$u(1, c, \theta') - u(1, c, \theta) \geq u(1, \bar{s}, \theta') - u(1, \bar{s}, \theta),$$

so the last display is weakly nonnegative. Hence

$$(1 - q) u^0(c) + q u(1, c, \theta') \geq (1 - q) u^0(0) + q u(1, \bar{s}, \theta').$$

Because $u^0(\cdot)$ and $u(1, \cdot, \theta')$ are decreasing, the map $s \mapsto (1 - q) u^0(s) + q u(1, s, \theta')$ is decreasing, and the solution $c(\theta')$ to (18) for type θ' must satisfy $c(\theta') \geq c(\theta)$. This proves WICE. $\square$

E.3 Comparison of Mechanisms

Building on the definitions and techniques from this paper, we provide a transparent comparison between non-coordination mechanisms and ex post certification mechanisms under the

weaker condition Assumption 6, which is satisfied by a large class of non-separable utilities (Lemma 2).

Proposition 3 (Non-coordination vs Certification). *Suppose Assumption 1, 2 and 6 hold. Fix any interim allocation–utility pair $(\mathbf{Q}, \mathbf{U})$ such that $(\mathbf{Q}, \mathbf{U})$ can be implemented by an ex post certification mechanism and Q_i is weakly increasing for all i . Then there exists an interim allocation–utility pair $(\mathbf{Q}^\dagger, \mathbf{U}^\dagger)$ that can be implemented by a direct non-coordination mechanism such that $\mathbf{Q}^\dagger = \mathbf{Q}$, and $U_i^\dagger(\theta_i) \geq U_i(\theta_i)$ for all i, θ_i .*

The proof of Proposition 3 is almost identical to the proof of Theorem 1, with the exception that, here, it suffices to consider distributions over signals and allocations that have binary support from each agent’s perspective. The details are omitted here to avoid repetition.

References

- Akbarpour, M., Dworczak, P., and Kominers, S. D. (2024). Redistributive allocation mechanisms. *Journal of Political Economy*, 132(6):1831–1875.
- Ball, I. (2024). Scoring strategic agents. *Accepted at American Economic Journal: Microeconomics*.
- Baye, M. R., Kovenock, D., and De Vries, C. G. (1996). The all-pay auction with complete information. *Economic Theory*, 8(2):291–305.
- Border, K. C. (1991). Implementation of reduced form auctions: A geometric approach. *Econometrica: Journal of the Econometric Society*, pages 1175–1187.
- Budish, E. (2011). The combinatorial assignment problem: Approximate competitive equilibrium from equal incomes. *Journal of Political Economy*, 119(6):1061–1103.
- Chakravarty, S. and Kaplan, T. R. (2013). Optimal allocation without transfer payments. *Games and Economic Behavior*, 77(1):1–20.
- Che, Y.-K. and Gale, I. (1998). Standard auctions with financially constrained bidders. *The Review of Economic Studies*, 65(1):1–21.
- Che, Y.-K., Kim, J., and Mierendorff, K. (2013). Generalized reduced-form auctions: A network-flow approach. *Econometrica*, 81(6):2487–2520.
- Condorelli, D. (2012). What money can’t buy: Efficient mechanism design with costly signals. *Games and Economic Behavior*, 75(2):613–624.
- Fang, D., Noe, T., and Strack, P. (2020). Turning up the heat: The discouraging effect of competition in contests. *Journal of Political Economy*, 128(5):1940–1975.
- Finkelstein, A. and Notowidigdo, M. J. (2019). Take-up and targeting: Experimental evidence from snap. *The Quarterly Journal of Economics*, 134(3):1505–1556.
- Frankel, A. and Kartik, N. (2019). Muddled information. *Journal of Political Economy*, 127(4):1739–1776.
- Gershkov, A., Moldovanu, B., Strack, P., and Zhang, M. (2022). Optimal insurance: Dual utility, random losses and adverse selection. *American Economic Review (submitted)*.
- Green, J. R. and Laffont, J.-J. (1986). Partially verifiable information and mechanism design. *Review of Economic Studies*, 53(3):447–456.

- Hardt, M., Megiddo, N., Papadimitriou, C., and Wootters, M. (2016). Strategic classification. In *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science*, pages 111–122.
- Hartline, J. D. and Roughgarden, T. (2008). Optimal mechanism design and money burning. In *Proceedings of the Fortieth Annual ACM Symposium on Theory of Computing*, pages 75–84.
- Kartik, N., Lee, S., and Rappoport, D. (2024). Single-crossing differences in convex environments. *Review of Economic Studies*, 91(5):2981–3012.
- Kazumura, T., Mishra, D., and Serizawa, S. (2020). Mechanism design without quasilinearity. *Theoretical Economics*, 15(2):511–544.
- Kim, K., Krishna, R. V., and Ryvkin, D. (2024). Choosing your own luck: Strategic risk taking and effort in contests. *Available at SSRN 4884945*.
- Kleiner, A., Moldovanu, B., and Strack, P. (2021). Extreme points and majorization: Economic applications. *Econometrica*, 89(4):1557–1593.
- Kos, N. and Messner, M. (2013). Incentive compatibility in non-quasilinear environments. *Economics Letters*, 121(1):12–14.
- Lazear, E. P. and Rosen, S. (1981). Rank-order tournaments as optimum labor contracts. *Journal of Political Economy*, 89(5):841–864.
- Li, Y. and Qiu, X. (2026). Allocating resources under strategic misrepresentation. *Working paper*.
- Maskin, E. and Riley, J. (1984). Optimal auctions with risk averse buyers. *Econometrica: Journal of the Econometric Society*, pages 1473–1518.
- Matthews, S. (1987). Comparing auctions for risk averse buyers: A buyer’s point of view. *Econometrica: Journal of the Econometric Society*, pages 633–646.
- McAfee, R. P. and McMillan, J. (1992). Bidding rings. *The American Economic Review*, pages 579–599.
- Milgrom, P. and Shannon, C. (1994). Monotone comparative statics. *Econometrica: Journal of the Econometric Society*, pages 157–180.

- Milgrom, P. R. and Weber, R. J. (1982). A theory of auctions and competitive bidding. *Econometrica: Journal of the Econometric Society*, pages 1089–1122.
- Moldovanu, B. and Sela, A. (2001). The optimal allocation of prizes in contests. *American Economic Review*, 91(3):542–558.
- Myerson, R. B. (1981). Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73.
- Myerson, R. B. (1982). Optimal coordination mechanisms in generalized principal–agent problems. *Journal of mathematical economics*, 10(1):67–81.
- Perez-Richet, E. and Skreta, V. (2022). Test design under falsification. *Econometrica*, 90(3):1109–1142.
- Perez-Richet, E. and Skreta, V. (2023). Fraud-proof non-market allocation mechanisms. *Working paper*.
- Perez-Richet, E. and Skreta, V. (2024). Score-based mechanisms. *arXiv preprint arXiv:2403.08031*.
- Pratt, J. W. (1976). Risk aversion in the small and in the large. *Econometrica*, 44(2):420.
- Quah, J. K.-H. and Strulovici, B. (2012). Aggregating the single crossing property. *Econometrica*, 80(5):2333–2348.
- Roth, A. E. (2002). The economist as engineer: Game theory, experimentation, and computation as tools for design economics. *Econometrica*, 70(4):1341–1378.
- Siegel, R. (2009). All-pay contests. *Econometrica*, 77(1):71–92.
- Skaperdas, S. (1996). Contest success functions. *Economic Theory*, 7(2):283–290.
- Spence, M. (1973). Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374.
- U.S. Government Accountability Office (2005). Hud rental assistance: Progress and challenges in measuring and reducing improper rent subsidies. Technical Report GAO-05-224, U.S. Government Accountability Office. Accessed June 9, 2025.
- Yang, F., Dworczak, P., and Akbarpour, M. (2024). Comparison of screening devices. *Revised & Resubmit, Journal of Political Economy*.
- Zhang, M. (2024). Optimal contests with incomplete information and convex effort costs. *Theoretical Economics*, 19(1):95–129.